

Extended Technical Appendix for “Topological-order-dependent axial advance of a local Stokes domain in Poincaré circular Airy beams”

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This appendix contains the extended derivations and full numerical comparisons. Its equation, figure, and table numbering is independent of the accompanying Supplementary Material.

1 Transverse polarization and the measured signal

The phasor convention is $\exp(-i\omega t)$ and $\mathbf{e}_\pm = (\mathbf{e}_x \pm i\mathbf{e}_y)/\sqrt{2}$. For $u = U_{m-P}$, $v = U_{m+P}$, the Cartesian field is

$$E_x = \frac{1}{2}(ue^{i(m-P)\theta} + ve^{i(m+P)\theta}), \quad E_y = \frac{i}{2}(ue^{i(m-P)\theta} - ve^{i(m+P)\theta}). \quad (\text{TA1})$$

Direct Jones algebra gives $S_0 = |E_x|^2 + |E_y|^2$, $S_1 = |E_x|^2 - |E_y|^2$, $S_2 = 2 \operatorname{Re}(E_x^* E_y)$, and $S_3 = 2 \operatorname{Im}(E_x^* E_y) = (|u|^2 - |v|^2)/2$. Thus, where $S_0 > 0$,

$$s_3 = \frac{|u|^2 - |v|^2}{|u|^2 + |v|^2} = \sin 2\chi, \quad (\text{TA2})$$

$$\psi = P\theta - \frac{\arg u - \arg v}{2} \pmod{\pi}.$$

The second expression is undefined at circular polarization. The orientation is obtained in the calculations from $\frac{1}{2} \operatorname{atan2}(S_2, S_1)$ and checked against the circular-channel expression. A common relative intensity mask $S_0/\max S_0 > 10^{-9}$ is used for the displayed maps, with 10^{-12} and 10^{-6} checked separately. The orientation also requires $\sqrt{S_1^2 + S_2^2}/S_0 > 10^{-8}$.

For an ideal linear analyzer at angle ϑ ,

$$I_\vartheta = \frac{1}{2}S_0[1 + V \cos\{2(\vartheta - P\theta) + \delta\}], \quad V = \frac{2|u||v|}{|u|^2 + |v|^2}, \quad \delta = \arg u - \arg v. \quad (\text{TA3})$$

For a resolved coherent transverse field, $V^2 + s_3^2 = 1$. Pixel, temporal, or ensemble averaging need not preserve this equality. A linear analyzer provides neither the sign of s_3 nor the individual circular-channel charges. For fixed $P = 1$, its twofold modulation is common to all m . Stokes polarimetry and circularly matched reference interferometry serve distinct purposes, as also illustrated in Supplementary Notes 6–7 of Liu *et al.*[1] and the detection discussion of Yan *et al.*[2]. Here the charge relation is $m = (q_+ + q_-)/2$, $P = (q_- - q_+)/2$.

Uniform transformations $u \mapsto ue^{i\phi_+}$ and $v \mapsto ve^{i\phi_-}$ leave S_3 exactly unchanged. They change ψ by $-(\phi_+ - \phi_-)/2$. This control does not remove the effect of spatial propagation phases on radial amplitudes. In the cached-field check at $R_0 = 24$, $\eta = 0.70$, $m = 1, 2, 4$, and common planes $\tau = -0.5, 0, 1$, the maximum normalized discrepancy between Cartesian and channel-based S_3 was 9.3×10^{-16} ; the uniform-phase control changed S_3 by less than 6.2×10^{-16} of the plane’s maximum S_0 .

For any common integration region,

$$Q = T\bar{p}_3, \quad T = \frac{P_+^{\text{ROI}} + P_-^{\text{ROI}}}{P_{\text{in}}}, \quad \bar{p}_3 = \frac{P_+^{\text{ROI}} - P_-^{\text{ROI}}}{P_+^{\text{ROI}} + P_-^{\text{ROI}}}. \quad (\text{TA4})$$

The factor T quantifies captured intensity; $\bar{p}_3$ quantifies its intensity-weighted circular imbalance. Neither is the arithmetic mean of χ . A vanishing denominator makes $\bar{p}_3$ undefined.

1.1 What Q adds to captured power

All Q , T , and $\bar{p}_3$ comparisons use the same propagated complex channels, input-power normalization, and ROI for each order. The Q and T events are nevertheless found independently from each signal's own first stationary maximum. Define

$$d_m = h_Q(m) - h_T(m), \quad E_m = \Delta_Q(m) - \Delta_T(m) = d_1 - d_m, \quad (\text{TA5})$$

where $\Delta_X(m) = h_X(1) - h_X(m)$. At $R_0 = 24$, $\eta = 0.7$, $m = 4$, $d_1 = 0.02626546$ and $d_4 = 0.02812356$, so the two order differences are close: $\Delta_Q = 0.29196941$, $\Delta_T = 0.29382751$, and $E_4 = -0.00185811$. This is a near-common-mode cancellation at that condition, not a decomposition of the advance into intensity and polarization percentages. At $R_0 = 96$ with the same η, m , $\Delta_Q = 0.07550782$, $\Delta_T = 0.09514515$, and $E_4 = -0.01963733$, or -26.0% of Δ_Q .

All 500 proxy comparisons for $m = 2-6$ have defined Q and T half-height events. Their median $|E_m|$ is 0.014207, the maximum is 0.250949, and the median $|E_m/\Delta_Q|$ is 0.13981. These summary values establish neither equivalence nor a universal dominance by one factor. The full machine-readable table includes each single-order event, d_m , both order differences, E_m , and status.

Without peak normalization, choose a common $\tau_* = 0$ and define

$$\mathcal{D}_X(\tau) = R_0 \ln \frac{X_m(\tau)X_1(\tau_*)}{X_1(\tau)X_m(\tau_*)}. \quad (\text{TA6})$$

The factorization gives the pointwise identity $\mathcal{D}_Q = \mathcal{D}_T + \mathcal{D}_{\bar{p}_3}$. Its maximum numerical error over all retained valid samples is below 7.4×10^{-14} . For $R_0 = 24$, $\eta = 0.7$, $m = 4$, the three slopes at $\tau = 0$ are -7.897934 , -7.721899 , and -0.176036 . At $R_0 = 96$ they are -8.962823 , -9.630689 , and $+0.667867$. The signed polarization factor therefore modifies amplitude and can also modify curve shape and event position.

Table TA1: Proxy-boundary comparison of Q and T events. Each signal uses its own first stationary maximum and rising half-height. Here $d_m = h_Q(m) - h_T(m)$, $\Delta_X = h_X(1) - h_X(m)$, and $E_m = \Delta_Q - \Delta_T = d_1 - d_m$.

R_0	η	m	d_1	d_m	Δ_Q	Δ_T	E_m	$100E_m/\Delta_Q$
24	0.35	2	+0.010670	+0.006233	0.083384	0.078947	+0.004437	+5.3
24	0.35	3	+0.010670	-0.008347	0.178589	0.159572	+0.019017	+10.6
24	0.35	4	+0.010670	-0.031406	0.284729	0.242653	+0.042076	+14.8
24	0.70	2	+0.026265	+0.035752	0.088964	0.098450	-0.009486	-10.7
24	0.70	3	+0.026265	+0.035600	0.186557	0.195891	-0.009334	-5.0
24	0.70	4	+0.026265	+0.028124	0.291969	0.293828	-0.001858	-0.6
96	0.35	2	+0.007388	+0.009786	0.021121	0.023519	-0.002398	-11.4
96	0.35	3	+0.007388	+0.009342	0.044790	0.046744	-0.001954	-4.4
96	0.35	4	+0.007388	+0.006855	0.070534	0.070000	+0.000534	+0.8
96	0.70	2	+0.017295	+0.027256	0.023127	0.033089	-0.009961	-43.1
96	0.70	3	+0.017295	+0.033372	0.048474	0.064551	-0.016077	-33.2
96	0.70	4	+0.017295	+0.036932	0.075508	0.095145	-0.019637	-26.0

1.2 Short-distance radial mechanism

For a real radial envelope $f(s)$, $s = \rho/w$, and $\zeta = z/(kw^2)$, the paraxial radial operator is $\mathcal{L}_q = \partial_s^2 + s^{-1}\partial_s - q^2s^{-2}$. At a smooth point away from the axis,

$$U_q = f + \frac{i\zeta}{2}\mathcal{L}_q f - \frac{\zeta^2}{8}\mathcal{L}_q^2 f + \dots \quad (\text{TA7})$$

With $V = s^{-2}$, $[\mathcal{D}, V]f = 4f/s^4 - 4f'/s^3$ for $\mathcal{D} = \partial_s^2 + s^{-1}\partial_s$. Hence

$$S_3(s, \zeta) = 2mP\zeta^2 \left(\frac{ff'}{s^3} - \frac{f^2}{s^4} \right) + O(\zeta^4). \quad (\text{TA8})$$

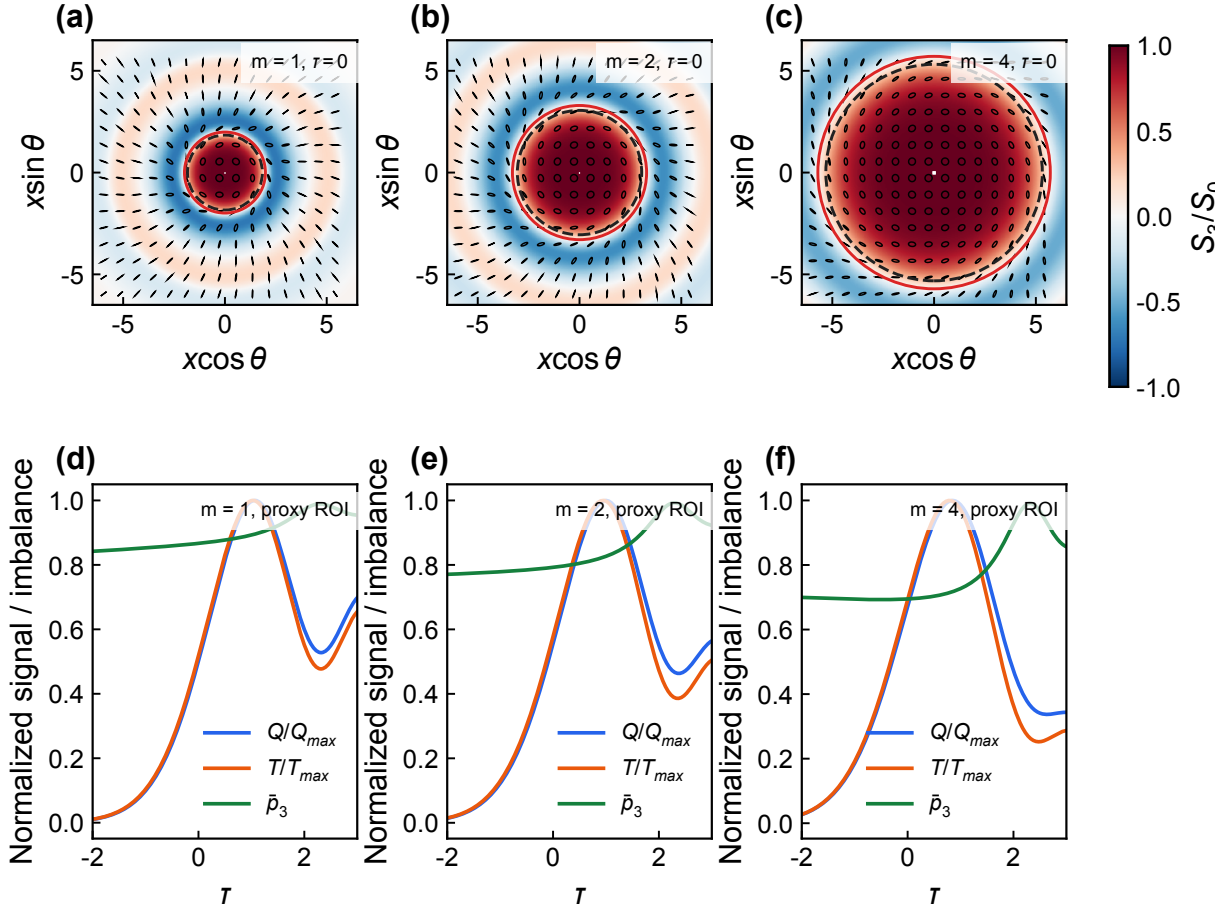


Figure TA1: Polarization and signal decomposition calculated from complex reduced R-S channels at $R_0 = 24$, $\eta = 0.70$. Top: common $\tau = 0$ planes, with $x = z\rho/(2kw^3)$. The color shows S_3/S_0 ; ellipses follow Cartesian Jones fields. Solid red circles mark the resolved actual $S_3 = 0$ contour; dashed circles mark $j'_{m,1}$. Bottom: Q and T separately normalized to their maxima over the stored interval, together with unscaled $\bar{p}_3$. The displayed interval is $-2 \leq \tau \leq 3$ and the stored interval is $[-3, 5]$. These traces separate captured intensity from polarization composition.

The order-dependent current difference at first order is $j_3 = -2mPzf^2/(k^2\rho^3)$ in physical coordinates. Its divergence produces the displayed second-order intensity imbalance. This is a local smooth-envelope expansion, not an expansion uniform at the axis or at the Airy caustic. It gives a direct connection between the radial phase gradient, current, and local circular imbalance. A uniform phase control removes none of these spatial gradients.

2 Moving integration regions

Let $b(z)$ denote a physical radius, distinct from a scaled upper limit. Leibniz differentiation at fixed physical ρ gives

$$\frac{dQ_b}{dz} = \frac{2\pi}{P_{\text{in}}} \left[\int_0^{b(z)} \partial_z S_3 \rho \, d\rho + b(z)b'(z)S_3(b(z), z) \right]. \quad (\text{TA9})$$

The physical proxy radius is $b(z) = 2kw^3 j'_{m,1}/z$, so $b' = -b/z$. A radius fixed at its value at z_c has $b' = 0$. An actual zero contour also has zero moving-limit term, because its integrand vanishes; its signal still differs from the proxy integral.

For lossless paraxial propagation, the radial channel currents are $j_{\pm} = \text{Im}(U_{\pm}^* \partial_{\rho} U_{\pm})/(2k)$. Writing $j_3 = j_+ - j_-$ gives

$$\frac{dQ_b}{dz} = \frac{2\pi b}{P_{\text{in}}} [-j_3(b, z) + b'S_3(b, z)]. \quad (\text{TA10})$$

The factor $1/2$ follows from the circular-channel amplitudes in the vector field. Equation (TA9) is a kinematic identity; Eq. (TA10) additionally assumes the paraxial wave equation. It is not a complete Maxwell spin-conservation law and is not assumed exact for the reduced R-S kernel. For $R_0 = 24$, $\eta = 0.70$, $m = 1, 2, 4$ and the predeclared interval $-0.5 \leq \tau \leq 1.25$, fixed-physical-radius differentiation, direct Leibniz integration, and the boundary-current expression were compared using independent Fresnel–Hankel fields. The separate terms are shown in Fig. TA2. The proxy has a nonzero moving-aperture contribution; the actual zero has a numerically vanishing one. This check tests the specified paraxial model and does not assert a full-field spin law.

3 Full-amplitude fold expansion

The main-text normalization uses $\mathcal{U}_q^{\text{in}} \sim (C_q/2\pi) \int_{\Gamma} \mathcal{F}_q e^{i(t^3/3+Xt)} dt$, where

$$C_q = E_0 \sqrt{2\pi R_0} (-i)^q e^{-i\pi/4} e^{\alpha(R_0 - \zeta^2/2)} \\ \times \exp\{i[kz + \zeta(R_0 + \alpha^2)/2 - \zeta^3/12]\}.$$

Here $s_c = R_0 + (u + i\alpha)^2$ and $g = \sqrt{s_c/R_0}$. The contour runs from $\infty e^{5\pi i/6}$ through zero to $\infty e^{\pi i/6}$, with moments $(2\pi)^{-1} \int_{\Gamma} t^n e^{i(t^3/3+Xt)} dt = (-i)^n \text{Ai}^{(n)}(X)$. The center $t = 0$ is the coalescence point at $X = 0$, not a stationary point for general X . The common z -dependent prefactor is retained throughout the peak and crossing calculation.

Put $\epsilon = R_0^{-1/2}$, $u_0 = \epsilon^{-1} + \epsilon(\tau - i\eta)/2$, and $X_0 = -\tau + i\eta$. With $u = u_0 + t$,

$$X = X_0 + \epsilon^2 X_2, \quad X_2 = -\tau^2/4 + i\eta\tau/2, \quad (\text{TA11})$$

$$\frac{u}{\sqrt{R_0}} = 1 + \epsilon t + \frac{1}{2}\epsilon^2(\tau - i\eta), \quad (\text{TA12})$$

$$\frac{s_c}{2R_0} = 1 + \epsilon t + \frac{1}{2}\epsilon^2(t^2 + \tau) + \frac{1}{2}\epsilon^3 t\tau + \frac{1}{8}\epsilon^4 \tau^2. \quad (\text{TA13})$$

The amplitude $g\sqrt{u}$ must be expanded together with $J_q(uR)$. After removal of a common nonzero factor, define

$$\ell_q = J_q + xJ'_q, \quad n_q = (q^2 - x^2 + \frac{1}{2})J_q + xJ'_q. \quad (\text{TA14})$$

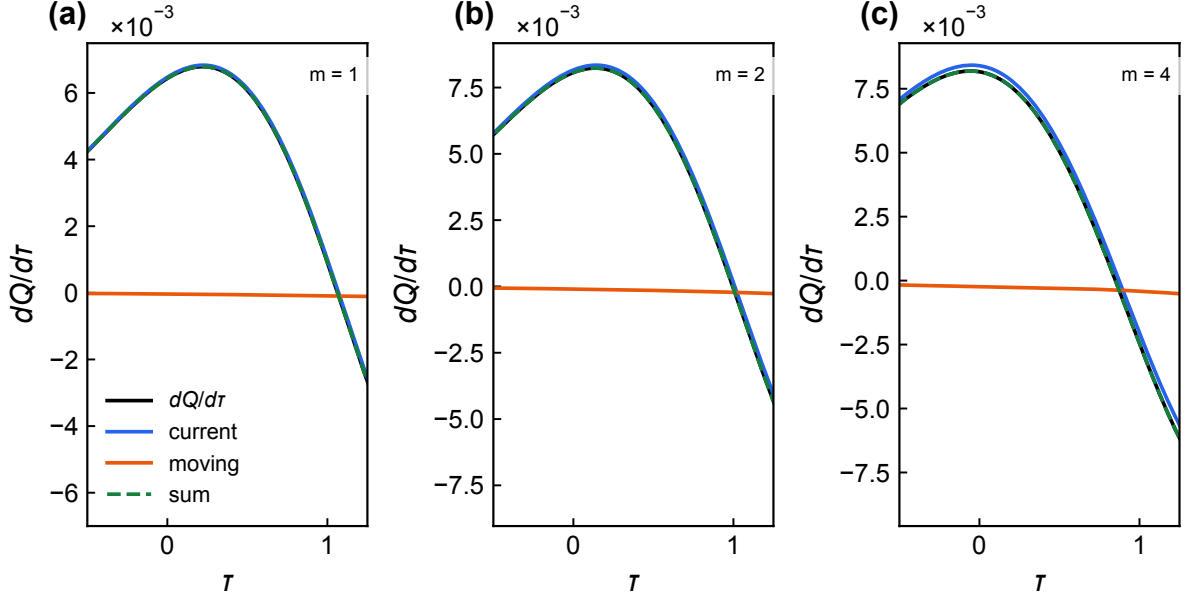


Figure TA2: Derivative of the proxy-domain signal and its paraxial radial-current and moving-aperture contributions. The dimensionless axis is $\tau = (z - z_c)/L_c$; plotted derivatives are $dQ/d\tau = L_c dQ/dz$. All panels use common physical propagation planes. The dashed sum is independently compared with differentiation of the integrated signal.

Writing $A = \text{Ai}(X_0)$ and using the Airy moments $t^j \mapsto (-i)^j \text{Ai}^{(j)}(X_0)$ gives

$$H_q = AJ_q - i\epsilon A' \ell_q \quad (\text{TA15})$$

$$+ \epsilon^2 \left[A \left\{ (\tau/2 - i\eta/4)J_q - i\eta x J'_q/2 \right\} - \frac{1}{2}A''n_q + X_2A'J_q \right] + O(\epsilon^3). \quad (\text{TA16})$$

The squared caustic prefactor contains $\exp[-2\eta\tau\epsilon - \eta\tau^2\epsilon^3/2]$ and the radial Jacobian contains $(1+\tau\epsilon^2/2)^{-2}$. A common multiplicative function of τ affects individual peaks and crossings and cannot be deleted before locating them.

At $b_0 = j'_{m,1}$, put $J_- = J_{m-1}$, $J_+ = J_{m+1}$ and $I_0 = \int_0^{b_0} x(J_-^2 - J_+^2) dx = 2mJ_m^2(b_0)$. The complete amplitude obeys

$$\int_0^{b_0} x(J_- \ell_- - J_+ \ell_+) dx = 0, \quad (\text{TA17})$$

$$\int_0^{b_0} x(\ell_-^2 - \ell_+^2) dx = 0, \quad (\text{TA18})$$

$$\int_0^{b_0} x(J_- n_- - J_+ n_+) dx = (D_m - \frac{1}{4})I_0. \quad (\text{TA19})$$

Together with $A'' = X_0A$, these identities yield

$$\frac{Q_m}{N_m} = f - 2\eta\tau f\epsilon + \epsilon^2 \left[(D_m - \frac{1}{4})\tau f + 2\eta^2\tau^2 f + 2 \text{Re}(A^* A' X_2) \right] + O(\epsilon^3), \quad (\text{TA20})$$

where $f = |A|^2$, $N_m = 4\pi^2|E_0|^2 w^2 \sqrt{R_0} e^{-\eta\sqrt{R_0}} I_0 / \mathcal{P}_{\text{in}}$, and $A' = \text{Ai}'(X_0)$ is an argument derivative, so $dA/d\tau = -A'$. The amplitude N_m is distinct from the fourth-order event coefficient C_m . All displayed terms except $D_m\tau f$ are common to different orders. The finite polynomial calculation through ϵ^4 retains the complex shift of u_0 , X_2 , derivatives of g , the first Debye correction, and every intensity cross product. It expands intensity consistently in ϵ ; truncating the field at derivative number N and then squaring is

a different operation. The opposite Hankel branch and the lower spectral endpoint are not included in this local fold series. Their contribution cannot be assigned to the series remainder without a separate comparison with the full propagated field.

For reproducibility, let the complete slowly varying amplitude before the Airy integral be the finite bivariate polynomial

$$P_q(\epsilon, t, x) = \sum_{n=0}^4 \sum_j p_{q,nj}(x) \epsilon^n t^j + O(\epsilon^5). \quad (\text{TA21})$$

The field coefficients used in the calculation are generated by

$$H_{q,n} = \sum_{r=0}^{\lfloor n/2 \rfloor} \sum_j p_{q,n-2r,j} \frac{X_2^r}{r!} (-i)^j \text{Ai}^{(j+r)}(X_0), \quad 0 \leq n \leq 4. \quad (\text{TA22})$$

This rule includes the $X = X_0 + \epsilon^2 X_2$ shift and the derivatives of the full amplitude. If the upper sign denotes T and the lower sign Q , the unprefactored intensity coefficient is

$$I_{m,n}^{T/Q}(x, \tau) = \sum_{a=0}^n \text{Re} [H_{m-1,a} H_{m-1,n-a}^* \pm H_{m+1,a} H_{m+1,n-a}^*]. \quad (\text{TA23})$$

It is integrated with the radial Jacobian and then convolved, order by order, with the common caustic-prefactor and moving-coordinate Jacobian series. Equation (TA23) keeps only products whose total order is n ; simply squaring a field truncated at a derivative count also creates terms above the requested intensity order.

Write the selected peak and rising crossing as

$$p_m = p_0 + \sum_{n=1}^4 p_{m,n} \epsilon^n, \quad h_m = h_0 + \sum_{n=1}^4 h_{m,n} \epsilon^n. \quad (\text{TA24})$$

For a signal $X_m(\tau, \epsilon) = \sum_{n=0}^4 x_{m,n}(\tau) \epsilon^n$, coefficient extraction from $X'_m(p_m) = 0$ and $X_m(h_m) = \gamma X_m(p_m)$ gives

$$p_{m,n} = - \frac{[\epsilon^n] X'_m(p_0 + \sum_{j<n} p_{m,j} \epsilon^j, \epsilon)}{x''_{m,0}(p_0)}, \quad (\text{TA25})$$

$$h_{m,n} = \frac{\gamma [\epsilon^n] X_m(p_0 + \sum_{j<n} p_{m,j} \epsilon^j, \epsilon) - [\epsilon^n] X_m(h_0 + \sum_{j<n} h_{m,j} \epsilon^j, \epsilon)}{x'_{m,0}(h_0)}. \quad (\text{TA26})$$

Here $[\epsilon^n]$ means coefficient extraction after substituting only previously known event coefficients. The apparently missing $p_{m,n}$ term in the peak value vanishes because $x'_{m,0}(p_0) = 0$. The difference $c_n = h_{1,n} - h_{m,n}$ gives

$$\Delta_X = c_1 R_0^{-1/2} + c_2 R_0^{-1} + c_3 R_0^{-3/2} + c_4 R_0^{-2} + O(R_0^{-5/2}). \quad (\text{TA27})$$

For the proxy Q event, $c_1 = 0$ analytically and $(A_m, B_m, C_m) = (c_2, c_3, c_4)$. For T , c_1 generally does not vanish. Table TA2 gives both observables.

The following command was run from the revision root and reads no propagation fit data:

```
PYTHONPATH=code /usr/bin/python3 code/generate_event_coefficients.py \
--m 4 --eta 0.7 --observable Q --gamma 0.5 --points 25 --nx 1201 \
--output theory/generated_coefficients_Q_m4_eta0.7.json
```

Replacing Q by T generated the corresponding T record. The $m = 4, \eta = 0.7$ Q result is $(c_1, c_2, c_3, c_4) = (9.2 \times 10^{-12}, 7.91778391, -8.32699328, 15.71031482)$; refinement changed any coefficient by at most 1.9×10^{-9} . For T the values are $(0.52165323, 3.37061776, 9.34708065, -45.57713040)$.

Table TA2: Non-fitted event-difference coefficients, $\Delta_X = c_1\epsilon + c_2\epsilon^2 + c_3\epsilon^3 + c_4\epsilon^4 + O(\epsilon^5)$, generated directly from the full-amplitude local fold series. The final column is the largest coefficient change when the derivative and radial quadratures are refined.

η	m	X	c_1	c_2	c_3	c_4	refinement
0.35	2	Q	1.51157×10^{-13}	2.072213	-0.450474	-0.093052	1.74×10^{-11}
0.35	3	Q	1.50768×10^{-12}	4.414678	-1.440168	2.587658	1.33×10^{-11}
0.35	4	Q	3.90316×10^{-12}	6.972064	-2.980978	9.165550	2.65×10^{-10}
0.35	2	T	0.0831185	1.387474	0.721991	-2.173510	9.68×10^{-12}
0.35	3	T	0.155601	2.852249	1.413683	-4.450309	5.38×10^{-11}
0.35	4	T	0.221353	4.386813	2.049090	-6.529970	9.61×10^{-11}
0.70	2	Q	3.4861×10^{-13}	2.353296	-1.446718	1.103438	8.79×10^{-10}
0.70	3	Q	3.53673×10^{-12}	5.013504	-4.216432	5.924855	1.16×10^{-9}
0.70	4	Q	9.2073×10^{-12}	7.917784	-8.326993	15.710315	1.86×10^{-9}
0.70	2	T	0.195882	1.034615	3.038342	-12.028248	2.66×10^{-8}
0.70	3	T	0.3667	2.163520	6.177498	-27.390268	7.17×10^{-8}
0.70	4	T	0.521653	3.370618	9.347081	-45.577130	1.30×10^{-7}
0.90	2	Q	6.05516×10^{-13}	2.896903	-6.047288	43.080367	1.35×10^{-8}
0.90	3	Q	6.01563×10^{-12}	6.171614	-14.812476	94.788905	1.47×10^{-9}
0.90	4	Q	1.56808×10^{-11}	9.746778	-26.230152	154.023590	2.30×10^{-8}
0.90	2	T	0.334402	-0.573780	31.524664	-638.159910	9.96×10^{-7}
0.90	3	T	0.626015	-1.057104	68.582793	-1550.863030	2.59×10^{-6}
0.90	4	T	0.890545	-1.481604	110.834418	-2759.205956	5.19×10^{-6}

4 General boundaries and general polarization index

For $a = |m - P|$, $c = |m + P|$, define $B = J_a^2 - J_c^2$ and

$$I_0(b) = \int_0^b xB(x) dx, \quad L(b) = \frac{\int_0^b x(\ell_a^2 - \ell_c^2) dx}{I_0(b)}, \quad N(b) = \frac{\int_0^b x(J_a n_a - J_c n_c) dx}{I_0(b)}. \quad (\text{TA28})$$

The complete first-order integral is $\int_0^b x(J_a \ell_a - J_c \ell_c) dx = b^2 B(b)/2$. For adjacent orders this reproduces the relative signal correction

$$2\epsilon \frac{\text{Im}(A^* A')}{f} \frac{bJ'_m(b)}{J_m(b)}. \quad (\text{TA29})$$

Near $b = b_0(1 + \delta_b)$, $bJ'_m(b)/J_m(b) = (m^2 - b_0^2)\delta_b + O(\delta_b^2)$. For a general boundary, both second-order integrals must also be retained. An integration by parts and the Bessel equation give, for each order q ,

$$\int_0^b x\ell_q^2 dx = b^3 J_q(b)J'_q(b) + \int_0^b x^3 J_q^2 dx + (1 - q^2) \int_0^b xJ_q^2 dx, \quad (\text{TA30})$$

$$\int_0^b xJ_q n_q dx = (q^2 - \frac{1}{2}) \int_0^b xJ_q^2 dx - \int_0^b x^3 J_q^2 dx + \frac{1}{2}b^2 J_q^2(b). \quad (\text{TA31})$$

For $P = 2, 3$, b_0 is independently defined as the first nontrivial positive-to-negative zero of B . It is generally not $j'_{m,1}$. At this root the first-order integral vanishes, but L generally does not. The order-dependent part of the proxy signal is therefore

$$\epsilon^2 \{N\tau f + L|A'|^2\}. \quad (\text{TA32})$$

For example, $(P, m) = (2, 2)$ gives $b_0 = 2.29991033$, $L = 1.74740368$, and $N = -1.78958753$; $(P, m) = (3, 3)$ gives $b_0 = 2.39837324$, $L = 2.22989242$, and $N = -1.76126354$. These coefficients were computed by direct Bessel integration and checked against the full-amplitude polynomial expansion before the new P -family propagation. They cannot be replaced by the $P = 1$ coefficient D_m . For $m = 0$ or $P = 0$, the two absolute channel orders are equal and their balanced radial intensities give identically zero transverse S_3 .

5 The moving actual zero

Write the normalized radial integrand as $H(x, \epsilon) = H_0(x) + \epsilon H_1(x) + \dots$, where $H_0 = xB$. Assume $H_0(b_0) = 0$ and $H'_0(b_0) < 0$. Then

$$b_s = b_0 + \epsilon b_1 + \dots, \quad b_1 = -\frac{H_1(b_0)}{H'_0(b_0)}, \quad (\text{TA33})$$

$$\int_0^{b_s} H \, dx - \int_0^{b_0} H \, dx = -\epsilon^2 \frac{H_1^2(b_0)}{2H'_0(b_0)} + O(\epsilon^3). \quad (\text{TA34})$$

Since $H_1(b_0) = \nu b_0 H'_0(b_0)$ with $\nu = \text{Im}(A'/A)$, the root displacement is $b_1 = -\nu b_0$. For adjacent orders, $H'_0(b_0) = 4m(m^2/b_0^2 - 1)J_m^2(b_0)$, and the integral correction divided by I_0 is $(3/4 - D_m)\nu^2\epsilon^2$. Consequently the actual-domain order dependence is $D_m[\tau - \nu(\tau)^2]/R_0$. These statements require the same simple root branch and $A \neq 0$. They do not apply through root loss, merger, or a change in the identity of the first sign-changing root.

6 Event response and conditioning

Let a positive signal have a stationary maximum p and preceding rising crossing h_γ satisfying $Q(h_\gamma) = \gamma Q(p)$. For a small functional perturbation,

$$\delta p = -\frac{\delta Q'(p)}{Q''(p)}, \quad \delta h_\gamma = \frac{\gamma \delta Q(p) - \delta Q(h_\gamma)}{Q'(h_\gamma)}. \quad (\text{TA35})$$

For a relative perturbation $\epsilon^2 dW(\tau)$ of f , the event coefficient is

$$K_\gamma[W] = \frac{\gamma f(p)[W(p) - W(h_\gamma)]}{f'(h_\gamma)}. \quad (\text{TA36})$$

The proxy uses $W = \tau$; the actual adjacent-order domain uses $W = \tau - \nu^2$. The general P proxy uses the two functions τ and $|A'/A|^2$, with coefficients N and L . For the peak, the proxy coefficient is $-f(p)/f''(p)$. For a positive-weight centroid over a fixed common window, the coefficient of a relative linear tilt is its variance in that window. A signed centroid or an almost vanishing integral has a different interpretation and can be ill conditioned.

In a statistical measurement, to first order,

$$\text{Var}(\hat{h}_\gamma) \simeq \frac{\gamma^2 \sigma_p^2 + \sigma_h^2 - 2\gamma \text{Cov}(\delta Q_p, \delta Q_h)}{[Q'(h_\gamma)]^2}. \quad (\text{TA37})$$

For $\hat{\Delta} = \hat{h}_1 - \hat{h}_m$, its variance additionally contains $-2 \text{Cov}(\hat{h}_1, \hat{h}_m)$. This linear formula does not describe branch switching or disappearing peaks; those events must remain in a full measurement-chain simulation.

7 Monotonicity and physical scaling paths

For $m > 0$, use the quadratic form

$$q_m[y] = \int_0^1 r |y'|^2 \, dr + m^2 \int_0^1 |y|^2 \frac{dr}{r} \quad (\text{TA38})$$

in $L^2((0, 1), r \, dr)$, on its finite-energy domain. On compact positive-order intervals this domain is common. The natural outer boundary is Neumann, $y'(1) = 0$, and the lowest regular mode is $y_m(r) = J_m(j'_{m,1}r)$.

Its eigenvalue is simple. At the origin $y_m = O(r^m)$, so the boundary term $ry_m y'_m$ vanishes and both Hellmann–Feynman integrals converge. Hence

$$\frac{d}{dm}(j'_{m,1})^2 = 2m \frac{\int_0^1 y_m^2 dr/r}{\int_0^1 r y_m^2 dr} > 2m, \quad (\text{TA39})$$

which proves $D'(m) < 0$. At $m = 0$ the constant Neumann mode has eigenvalue zero, while the first positive derivative zero follows a different indexing convention; it must not be substituted into this proof.

At fixed k, w, η , the leading dimensionful advance is

$$\Delta z = \frac{k w^2}{\sqrt{R_0}} \Delta \tau \sim k w^2 K_\eta (D_1 - D_m) R_0^{-3/2}. \quad (\text{TA40})$$

Holding α fixed instead gives $\eta = 2\alpha\sqrt{R_0}$ and changes the axial profile and its event coefficient. When $k w$ is varied at fixed wavelength, $w = (k w)/k$ must be varied consistently. The accumulated nonparaxial phase depends on the occupied angular spectrum and propagation distance, not only on the maximum angle.

8 Numerical propagation and normalization

All dimensional calculations use millimeters internally, with $\lambda = 0.0006328$ mm and baseline $w = 0.08$ mm, hence $k w = 794.3344257$. The envelope amplitude is arbitrary because signals are normalized by $\mathcal{P}_{\text{in}}$. The original 36 beams use $m = 1, 2, 3$, $\eta = 0.35, 0.7$, and $R_0 = 24, 32, 40, 56, 72, 96$. The archived calculation and current implementation agree in all 432 stored comparisons at their saved precision; numerical convergence is assessed separately below.

8.1 Independent implementations

The source-radius-expanded R-S integral is evaluated with composite Simpson quadrature in $s = r/w$. A kernel coordinate y is defined by $a = y/(2R_0)$, $c = [1 - (a/kw)^2]^{1/2}$, $\rho = za/(kwc)$. Its relation to the physical scaled radius is $x = z\rho/(2kw^3)$; y is only a computational coordinate. Bessel functions are evaluated directly for each required integer order. The source phase is split into a separable part and a small correction containing $c - 1$. Four terms in this correction are retained, with an absolute exponential-remainder bound evaluated for every grid. The largest stored bound for the tail correction in the main matrix is 4.18×10^{-14} in field amplitude; it is not a bound on the event error, which also depends on signal conditioning.

The independent spectral calculation forms

$$H_q(u) = \int_0^\infty s f(s) J_{|q|}(us) ds, \quad U_q(r, \zeta) = \int_0^{u_{\max}} u H_q(u) J_{|q|}(ur) e^{i\zeta\Omega(u)} du, \quad (\text{TA41})$$

where $r = \rho/w$ and the carrier is removed. The rates are $\Omega_F = -u^2/2$ and

$$\Omega_{\text{exact}} = k w \{ [(k w)^2 - u^2]^{1/2} - k w \} = - \frac{u^2 k w}{[(k w)^2 - u^2]^{1/2} + k w}. \quad (\text{TA42})$$

The forward transform uses a logarithmic Hankel grid on $10^{-6} \leq s \leq 10^4$, with 2^{21} points below $R_0 = 192$ and 2^{22} points at larger radius. The logarithmic increment is computed from the exact endpoint logarithms, avoiding cancellation in a ratio of adjacent nearly equal grid points. The inverse transform uses uniform spectral Simpson integration, including the independently integrated $u = 0$ value for $q = 0$, rather than a periodic inverse logarithmic transform. The retained limit is $u_{\max} = \sqrt{24/\alpha}$, below the grazing region in these comparisons. Production spectral increments are at most 0.00125, 0.000625, and 0.0003125 for small, intermediate, and large radii, respectively.

An independent spatial calculation uses the unexpanded first R-S kernel

$$K(d) = Z \frac{e^{ikw(d-Z)}}{d^2} \left(\frac{1}{d} - ikw \right), \quad d^2 = Z^2 + s^2 + r^2 - 2sr \cos \varphi, \quad Z = z/w. \quad (\text{TA43})$$

Angular integration projects the required charge, and radial integration uses an independent Simpson grid. At $\tau = -0.5, 0, 1$ and $x = 0, 1, 3, 6$, complex fields are compared on all 17 spectral parameter sets. The largest relative L^2 discrepancy is 4.86×10^{-8} . This agreement concerns the same scalar boundary problem. Reduced R-S, Fresnel, and exact propagation are model comparisons; exact spectral versus unexpanded R-S differences are implementation comparisons.

8.2 Extent, sampling, and event convergence

The production source grid has $\Delta s \leq 0.025$ and $s_{\max} = R_0 + \max(45, c/\alpha)$ with $c = \lceil 16 + \alpha R_0 \rceil$. A smaller preliminary cutoff was found insufficient at large R_0 and strong apodization. Its omitted tail was added to the cached channels as a complex amplitude; intensities were then recomputed including interference. Old data are retained separately. Full-source versus split-tail calculations at three representative cases change event positions by less than 8×10^{-9} .

Independent convergence axes include source increments 0.04, 0.025, 0.0125; observation counts 301, 601, and 1201 for source quadrature and 201, 401, and 801 for spectral propagation; and axial steps 0.05, 0.025, and 0.0125 when assessed on the available nested samples. The spectral forward transform is tested at $2^{21}, 2^{22}, 2^{23}$ points and its inverse increment at 0.000625, 0.0003125, 0.00015625 at $R_0 = 192$, $\eta = 0.2$. Additional source tails increase c by 2 and 4 at (24, 0.7), (192, 0.2), (256, 0.9), (256, 1.0). Spectral limits $\sqrt{20/\alpha}$, $\sqrt{24/\alpha}$, and $\sqrt{28/\alpha}$ are compared by coherent annular corrections, preserving the stored core field. Each refinement changes one numerical axis at a time. The two annular spectral-extent changes alter individual half-heights by at most 6.92×10^{-9} over the three checked parameter sets.

The final source-tail, spectral-sampling, observation, and axial refinements give maximum half-height and advance changes of 1.47×10^{-8} and 1.88×10^{-8} in the tested set. A cubic-to-quintic locator comparison at 27 representative beam cases gives a maximum half-height change 1.14×10^{-8} ; stationary-peak changes reach 9.87×10^{-7} , demonstrating that crossing and peak accuracies are different. A Brent tolerance is not a total numerical error estimate. All detailed values, including earlier failed convergence levels, are provided in the source-data tables.

The original two-grid advance change, 4.14×10^{-7} , omitted source extent as an independent axis. Extending that extent changes original single-order positions by up to 1.45×10^{-5} and advances by 1.71×10^{-5} . Original and revised values are tabulated side by side. Consequently the former grid estimate is not used as the uncertainty of the revised calculation. Power convergence alone was also insufficient to establish event convergence.

9 Boundary continuation and event records

Each saved complex field supplies 17 regions: 13 fixed scaled factors 0.8, 0.85, 0.9, 0.925, 0.95, 0.975, 1, 1.025, 1.05, 1.075, 1.1, 1.15, 1.2; the first actual positive-to-negative S_3 zero; a physical radius frozen at the m -dependent proxy value at z_c ; one common physical radius given by the $m = 1$ proxy at z_c ; and a raised-cosine soft window. The soft window is unity below $0.95b_0$, zero above $1.05b_0$, and transitions continuously between them. Channel powers, their sum and difference, actual roots, and negative- S_3 contributions are all saved. The identity $Q = T\bar{p}_3$ is evaluated on the same region and input normalization; it cannot compare regions with different denominators.

To evaluate boundary sensitivity, we compare $0.9b_0$, b_0 , $1.1b_0$, and the first actual positive-to-negative zero. For every boundary and every order, the first stationary maximum and preceding rising half-height are reselected from that boundary's own signal. The $m = 1$ reference uses the same construction rule as the target order. Table TA4 covers all six original radii, both original apodizations, and $m = 2, 3, 4$. It reports signed advances, absolute changes from b_0 , signed relative changes, and actual-root event status. The absolute difference remains the primary quantity when the reference advance is small.

Table TA3: Different error questions, with representative values in dimensionless τ . Values from distinct rows are not summed.

Quantity	Comparison and scope	Value
Numerical half-height change	Last isolated representative refinements	$\leq 1.47 \times 10^{-8}$
Numerical advance change	Same refinement set	$\leq 1.88 \times 10^{-8}$
Source-extent correction	Original 36 beams; maximum advance change	1.71×10^{-5}
Single-event model difference	Exact minus reduced; $R_0 = 192, \eta = 0.2, m = 4$	-0.05850
Advance model difference	Same beam and reference $m = 1$	-6.92×10^{-6}
Fresnel minus reduced advance	Same beam; source-radius/paraxial comparison	-4.58×10^{-9}
Exact minus Fresnel advance	Same beam; exact longitudinal wavenumber	-6.92×10^{-6}
Boundary-definition change	+10% radius; $R_0 = 24, \eta = 0.7, m = 4$	$+0.009368$
Formal-series residual	$\Delta - (A/R_0 + B/R_0^{3/2} + C/R_0^2); R_0 = 256, \eta = 0.7, m = 4$	1.70×10^{-5}

The postprocessing stores separate statuses for the Q pair, the T pair, their joint comparison, and continuity of the stored selected-root sequence. Among the actual-root half-height comparisons, 46 have a defined Q order difference but no joint Q/T comparison because a required T crossing is unavailable. These records remain valid for Q-only boundary tables and are not silently reclassified as missing Q events. Selected-root interruptions are reported separately from event existence.

The wider $0.8b_0$ – $1.2b_0$ family is a stress test, not a replacement protocol. At $R_0 = 96, \eta = 0.7$, the $1.2b_0$ advances are $-0.006342, -0.019772$, and -0.059266 for $m = 2, 3, 4$, respectively, whereas the corresponding b_0 events remain positive. This reversal is retained. It cannot be removed by narrowing the axial window or by calling the changed aperture a numerical error.

The continuity label is a diagnostic of the stored selected-root sequence. In the main-text display, adjacent finite roots are connected only when their separation is at most 0.3 in x . A missing root or a larger jump interrupts that sequence; this does not prove a branch exchange or a discontinuity of the propagated field. Identifying individual branches through such intervals would require separate root-continuation data.

Table TA4: Four-boundary half-height advances for the original six radii.

R_0	η	m	$0.9b_0$	b_0	$1.1b_0$	actual zero	status
			0.07356	0.08338	0.09602	0.08677	
24	0.35	2	(0.00983, -11.8%)	(0.00000, $+0.0\%$)	(0.01264, $+15.2\%$)	(0.00339, $+4.1\%$)	BR
			0.14966	0.17859	0.21896	0.18733	
24	0.35	3	(0.02893, -16.2%)	(0.00000, $+0.0\%$)	(0.04037, $+22.6\%$)	(0.00874, $+4.9\%$)	BR
			0.22740	0.28473	0.37116	0.30042	
24	0.35	4	(0.05732, -20.1%)	(0.00000, $+0.0\%$)	(0.08643, $+30.4\%$)	(0.01569, $+5.5\%$)	BR
			0.09082	0.08896	0.08346	0.08429	
24	0.70	2	(0.00186, $+2.1\%$)	(0.00000, $+0.0\%$)	(0.00551, -6.2%)	(0.00467, -5.2%)	D
			0.18232	0.18656	0.18377	0.18517	
24	0.70	3	(0.00423, -2.3%)	(0.00000, $+0.0\%$)	(0.00278, -1.5%)	(0.00139, -0.7%)	BR
			0.27369	0.29197	0.30134	0.30218	
24	0.70	4	(0.01828, -6.3%)	(0.00000, $+0.0\%$)	(0.00937, $+3.2\%$)	(0.01021, $+3.5\%$)	BR
			0.05671	0.06265	0.07004	0.06464	
32	0.35	2	(0.00594, -9.5%)	(0.00000, $+0.0\%$)	(0.00739, $+11.8\%$)	(0.00199, $+3.2\%$)	D
			0.11505	0.13361	0.15876	0.13913	
32	0.35	3	(0.01855, -13.9%)	(0.00000, $+0.0\%$)	(0.02515, $+18.8\%$)	(0.00552, $+4.1\%$)	BR
			0.17414	0.21199	0.26759	0.22216	
32	0.35	4	(0.03785, -17.9%)	(0.00000, $+0.0\%$)	(0.05560, $+26.2\%$)	(0.01017, $+4.8\%$)	BR
			0.07168	0.06722	0.05867	0.06065	
32	0.70	2	(0.00446, $+6.6\%$)	(0.00000, $+0.0\%$)	(0.00856, -12.7%)	(0.00657, -9.8%)	D

Table TA4 continued

R_0	η	m	$0.9b_0$	b_0	$1.1b_0$	actual zero	status
			0.14382	0.14067	0.12855	0.13230	
32	0.70	3	(0.00314, +2.2%)	(0.00000, +0.0%)	(0.01212, -8.6%)	(0.00838, -6.0%)	D
			0.21550	0.21943	0.20929	0.21400	
32	0.70	4	(0.00393, -1.8%)	(0.00000, +0.0%)	(0.01015, -4.6%)	(0.00543, -2.5%)	BR
			0.04648	0.05022	0.05470	0.05144	
40	0.35	2	(0.00375, -7.5%)	(0.00000, +0.0%)	(0.00447, +8.9%)	(0.00121, +2.4%)	D
			0.09419	0.10687	0.12359	0.11056	
40	0.35	3	(0.01269, -11.9%)	(0.00000, +0.0%)	(0.01672, +15.6%)	(0.00369, +3.5%)	BR
			0.14231	0.16912	0.20755	0.17621	
40	0.35	4	(0.02682, -15.9%)	(0.00000, +0.0%)	(0.03842, +22.7%)	(0.00709, +4.2%)	BR
			0.05991	0.05413	0.04413	0.04712	
40	0.70	2	(0.00577, +10.7%)	(0.00000, +0.0%)	(0.01000, -18.5%)	(0.00701, -13.0%)	D
			0.12027	0.11321	0.09645	0.10240	
40	0.70	3	(0.00706, +6.2%)	(0.00000, +0.0%)	(0.01676, -14.8%)	(0.01082, -9.6%)	D
			0.18018	0.17636	0.15638	0.16493	
40	0.70	4	(0.00382, +2.2%)	(0.00000, +0.0%)	(0.01998, -11.3%)	(0.01144, -6.5%)	D
			0.03458	0.03600	0.03744	0.03644	
56	0.35	2	(0.00142, -4.0%)	(0.00000, +0.0%)	(0.00144, +4.0%)	(0.00044, +1.2%)	D
			0.07004	0.07645	0.08434	0.07821	
56	0.35	3	(0.00641, -8.4%)	(0.00000, +0.0%)	(0.00788, +10.3%)	(0.00176, +2.3%)	BR
			0.10569	0.12066	0.14105	0.12444	
56	0.35	4	(0.01497, -12.4%)	(0.00000, +0.0%)	(0.02039, +16.9%)	(0.00378, +3.1%)	BR
			0.04599	0.03906	0.02794	0.03232	
56	0.70	2	(0.00693, +17.8%)	(0.00000, +0.0%)	(0.01111, -28.5%)	(0.00674, -17.3%)	D
			0.09251	0.08171	0.06094	0.06996	
56	0.70	3	(0.01081, +13.2%)	(0.00000, +0.0%)	(0.02077, -25.4%)	(0.01175, -14.4%)	D
			0.13873	0.12719	0.09833	0.11220	
56	0.70	4	(0.01154, +9.1%)	(0.00000, +0.0%)	(0.02885, -22.7%)	(0.01499, -11.8%)	D
			0.02782	0.02808	0.02802	0.02816	
72	0.35	2	(0.00026, -0.9%)	(0.00000, +0.0%)	(0.00006, -0.2%)	(0.00008, +0.3%)	D
			0.05637	0.05957	0.06304	0.06040	
72	0.35	3	(0.00321, -5.4%)	(0.00000, +0.0%)	(0.00347, +5.8%)	(0.00083, +1.4%)	D
			0.08504	0.09390	0.10523	0.09603	
72	0.35	4	(0.00886, -9.4%)	(0.00000, +0.0%)	(0.01133, +12.1%)	(0.00213, +2.3%)	BR
			0.03792	0.03060	0.01925	0.02445	
72	0.70	2	(0.00732, +23.9%)	(0.00000, +0.0%)	(0.01135, -37.1%)	(0.00615, -20.1%)	D
			0.07643	0.06406	0.04194	0.05281	
72	0.70	3	(0.01237, +19.3%)	(0.00000, +0.0%)	(0.02212, -34.5%)	(0.01125, -17.6%)	D
			0.11477	0.09973	0.06749	0.08451	
72	0.70	4	(0.01504, +15.1%)	(0.00000, +0.0%)	(0.03224, -32.3%)	(0.01522, -15.3%)	D
			0.02177	0.02112	0.01992	0.02097	
96	0.35	2	(0.00065, +3.1%)	(0.00000, +0.0%)	(0.00120, -5.7%)	(0.00015, -0.7%)	D
			0.04414	0.04479	0.04481	0.04494	
96	0.35	3	(0.00065, -1.4%)	(0.00000, +0.0%)	(0.00002, +0.0%)	(0.00015, +0.3%)	D
			0.06662	0.07053	0.07471	0.07140	
96	0.35	4	(0.00391, -5.5%)	(0.00000, +0.0%)	(0.00418, +5.9%)	(0.00086, +1.2%)	BR
			0.03055	0.02313	0.01193	0.01780	
96	0.70	2	(0.00742, +32.1%)	(0.00000, +0.0%)	(0.01120, -48.4%)	(0.00532, -23.0%)	D
			0.06173	0.04847	0.02598	0.03839	
96	0.70	3	(0.01325, +27.3%)	(0.00000, +0.0%)	(0.02249, -46.4%)	(0.01009, -20.8%)	D
			0.09287	0.07551	0.04168	0.06131	
96	0.70	4	(0.01736, +23.0%)	(0.00000, +0.0%)	(0.03383, -44.8%)	(0.01420, -18.8%)	D

Each cell is $\Delta\tau_{50}$ on the first line and $(|\Delta - \Delta_{b_0}|, 100(\Delta - \Delta_{b_0})/\Delta_{b_0})$ on the second line. D denotes a defined event and BR a defined event whose stored selected-zero sequence contains an interruption; such cases are not used as one smooth asymptotic branch.

Every positive-to-negative root candidate is retained. The trivial vortex axis zero is excluded. Initial processing uses a relative S_3 floor 10^{-10} and a first-cell exclusion; at four critical cases an independent axis moment determines $U_q/x^{|q|}$ as $x \rightarrow 0$. Spline interpolation of this nonsingular quantity in x^2 resolves the

root equation without an intensity floor. It detects no additional root below the original first radial cell in those cases. This is a numerical test at stated points, not a proof that every dark root is resolvable.

Root jumps larger than 0.3 in x break the initial event segments. Continuation identifiers also match all candidates to their linear predictions between planes; maximum matching distances 0.15, 0.3, 0.6 are audited separately. The 1800 records have unchanged event statuses and half-height differences below 7×10^{-16} relative to the initial segmentation. These identifiers label numerical continuations and are not topological invariants. No spline bridges an absent root or an interrupted branch. The selected event is the earliest interior positive maximum on a valid segment; all other extrema and crossing candidates remain in the archive. If the first segment has no valid peak, later valid segments are examined explicitly and their interval is reported.

An adaptive extent calculation expands the kernel coordinate to $y = 32$ and the axial interval to $[-8, 12]$ for three $R_0 = 24$ cases with $\eta = 0.35, 0.7, 0.9$, four $R_0 = 96$ cases with $\eta = 0.85, 0.9, 0.925, 0.95$, and $(R_0, \eta) = (256, 1.0)$. Cached inner-field planes are reused; only new axial planes and the outer annulus are propagated. The wider search does not supply a universal formation event. In particular, the actual-root $m = 4, 6$ events remain unavailable on their valid branches for the tested $R_0 = 96, \eta = 0.9, 0.925, 0.95$ cases, while $\eta = 0.85$ has defined selected events. Failure of a search and nonexistence on a particular continuous branch are recorded separately from propagation failure.

For $\eta = 0.7, m = 4$, the formal leading coefficient multiplying $(D_1 - D_4)/R_0$ is 0.80087525 for the proxy and 0.45152034 for the actual-zero boundary. The finite-radius advances can nevertheless be close: at $R_0 = 24$ they are 0.291969 and 0.302175. Their subleading terms are still large, and the actual-root record at that radius contains selected-root interruptions. On the uninterrupted large-radius branch, the effective coefficients $R_0\Delta/(D_1 - D_4)$ reach 0.75488 for the proxy and 0.54337 for the actual zero at $R_0 = 256$, moving toward distinct limits. At $R_0 = 16$, the actual-root Q pair is defined with $\Delta_Q = 0.071292$, but its root records contain selected-root interruptions; the target-order T rising crossing is absent. The Q point is therefore shown as a selected-root-interruption point and excluded from the smooth Q convergence sequence, while the joint Q/T comparison is undefined. The underlying records are in `actual_proxy_asymptotic_comparison.csv`.

Rising fractions 0.1–0.9 are evaluated relative to each region’s own first peak. Same-peak FWHM requires the descending half-height crossing to precede the next local minimum. Among the 10200 main-matrix region records, 6530 widths meet this condition, 278 cross half-height only beyond the next minimum, and 3392 lack a pair of crossings. The latter two groups remain undefined for same-peak width. The earlier, less restrictive crossing separation is retained as a separate diagnostic. Common-window centroids use $[-1, 2]$ for every order and require a continuous defined region there; signed weighting and nearly vanishing integrals are flagged. First-peak conditioning is reported as $Q(p)/|Q''(p)|$ and half-height conditioning as $Q(p)/|Q'(h)|$. They measure sensitivity to specified perturbations, not detector error without a noise model.

Full-curve registration fits a shift and an amplitude over $[-1, 2]$ with a shift search $[-1.9, 1.9]$ and reports the remaining relative L^2 shape error. Its interpolation domain stays within the saved $[-3, 5]$ interval. Double ratios instead use $\tau_* = 0$ and $\tau \in [-0.5, 0.5]$, requiring all four values above 10^{-8} of the larger peak. A measured regression slope is retained only as a diagnostic; the analytic slope is never replaced by it.

10 Higher-order, endpoint, and physical-path checks

The analytic A_m, B_m, C_m coefficients follow from polynomial algebra through ϵ^4 and implicit perturbation of the peak and crossing. All products of lower-order field terms are retained. The next radial saddle-amplitude correction can be checked independently using the tilted Airy moment: with $a = \alpha - iu$ and $s_* = R_0 - a^2$, it is $-a/(4s_*^2) = i\epsilon^3/16 - i\epsilon^4 t/16 + \dots$. The ϵ^3 part is a common phase, and the normalized intensity correction at ϵ^4 is $-\text{Re}(A^* A')/8$. It changes individual fourth-order events but cancels from the fourth-order order difference. Algebraic checks across orders give agreement to 2.8×10^{-13} . This check does not fit the generated $m = 4$ difference coefficients to propagation data.

The original derivative-number truncation through $N = 4$ is retained as a baseline truncation comparison. It is not identical to a power series truncated consistently in ϵ . Its original maximum single-event discrepancy was 1.639×10^{-3} , whereas its representative $m = 3$ advance discrepancy was 1.756×10^{-3} . Those are approximation errors, not quadrature errors. The four original figures and 36 parameter cases define the baseline comparison set.

The formal coefficients are compared without fitting on all available main-grid radii. Separate least-squares diagnostics fix A_m and fit B_m, C_m using $R_0 \leq 96$; the held-out radii are 128, 192, 256. The fit coefficients are labelled empirical and kept separate from formal coefficients. Near a degenerate Airy maximum, large higher-order coefficients and finite-radius peaks do not validate a nondegenerate asymptotic event. At $\eta = 1$, the finite-radius proxy events are retained but no continuation of the simple leading Airy event is asserted.

The fourth-order series at $\eta = 0.35$ does not uniformly improve the smallest radii and does not locate the intermediate-radius minimum. The focused calculation in Sec. 14 now closes this residual for $m = 3$: a compact Airy identity reproduces the full propagated signal, and its non-fitted expansion through fourteenth order predicts the minimum at $R_0 = 36.37128$, compared with 36.37130 for reduced R-S propagation. The missing fifth- and sixth-order coefficients contain both field products and nonlinear event terms. This result supersedes the earlier unresolved attribution for the fixed-proxy $m = 3$ event; it does not turn a partition-dependent endpoint or Hankel component into an independently measurable power. The earlier $m = 4$ and other-parameter comparisons remain finite-order diagnostics with their stated scope.

To examine omitted spectral contributions, the source-axis neighborhood is smoothly removed from individual Hankel branches below $R_0 - 10$ and restored through a transition ending at $R_0 - 8$. A high-frequency partition is zero below $u_a = \sqrt{R_0}/d$ and one above $2u_a$, with raised-cosine transition and $d = 6, 4, 3$. The inward and outward Hankel transforms are propagated coherently; $(1 - \chi)H_q$ defines the low-frequency endpoint component. The remaining high-frequency field is retained as an explicit remainder. The Debye transform with 8 and 12 terms is checked against direct Hankel-function integration. Individual components depend on the chosen partition, so their squared norms are not additive powers and cannot uniquely allocate a physical mechanism to the residual.

For $\eta = 0.7$, $m = 4$, $R_0 = 24$, removing the endpoint component changes the advance by between approximately -0.0239 and $+0.0132$ across the three partitions. At $R_0 = 256$, the corresponding changes are between -4.79×10^{-5} and $+1.04 \times 10^{-5}$. The outward component is smaller in field norm at these cases, but cannot be ignored merely because it has no inward stationary point. The full coherent sum is partition independent to the implementation accuracy. Additional $\eta = 0.35$ calculations at $R_0 = 24, 32, 40, 56$ test the nonmonotonic-residual region without changing its data or analytic coefficients.

The 61 extended parameter jobs contain 276 beams: 40 high-order controls including $m = 1, 8, 12, 16$; 16 small-ring cases at $R_0 = 8, 12$; 88 generalized- P beams; 60 fixed- α beams; and 72 varying- $k w$ beams. High-order searches use a wider radial observation range and $-6 \leq \tau \leq 8$. At fixed $\alpha = 0.02, 0.04, 0.07$, η varies with R_0 rather than remaining on the principal asymptotic path. Twenty fixed- α beams have no selected proxy peak in their stated search interval. They remain in the tables. No additional validity range is inferred merely from a successful run.

For $P = 2, 3$, let h_P be the reference event with $m = P$. The derived leading difference is

$$h_P - h_m = \frac{K_\gamma[\tau](N_{P,P} - N_{P,m}) + K_\gamma[|A'/A|^2](L_{P,P} - L_{P,m})}{R_0} + \dots \quad (\text{TA44})$$

At $\eta = 0.7$, $P = 2$, $m = 4$, its coefficient is 4.61595486; the computed difference at $R_0 = 192$ is 0.02490738, compared with the leading 0.02404143. For $P = 3$, $m = 5$, the coefficient is 4.30569959 and the corresponding values are 0.02477906 and 0.02242552. These finite-radius discrepancies are retained. The $m < P$ controls also have positive advances relative to $m = P$ in the sampled set, so monotonicity with m from the adjacent-order $P = 1$ proof is not a general- P statement.

11 Maxwell completion and longitudinal diagnostics

Let $a = m - 1$, $b = m + 1$, $\kappa = \sqrt{(kw)^2 - u^2}$ and suppress the common propagation exponential in the integrals below. Preserving the specified transverse spectrum gives

$$E_z(r, \theta) = \frac{ie^{im\theta}}{2} \int u^2 \frac{H_b - H_a}{\kappa} J_m(ur) e^{i\zeta\Omega} du, \quad (\text{TA45})$$

$$V_a(r) = \int u \left[\frac{\kappa}{kw} H_a + \frac{u^2(H_a - H_b)}{2kw\kappa} \right] J_a(ur) e^{i\zeta\Omega} du, \quad (\text{TA46})$$

with V_b obtained by interchanging a, b . Writing magnetic fields in impedance-scaled units,

$$Z_0 H_x = \frac{i}{2} (V_b e^{ib\theta} - V_a e^{ia\theta}), \quad Z_0 H_y = \frac{1}{2} (V_a e^{ia\theta} + V_b e^{ib\theta}), \quad (\text{TA47})$$

$$Z_0 H_z = -\frac{e^{im\theta}}{2kw} \int u^2 (H_a + H_b) J_m(ur) e^{i\zeta\Omega} du. \quad (\text{TA48})$$

The axial Poynting flux obeys $2Z_0 \langle S_z \rangle = \text{Re}(U_a V_a^* + U_b V_b^*)/2$. This completion differs from projecting a guessed three-component input and thereby changing the prescribed transverse channels.

The 27 beam/waist cases combine $(R_0, \eta) = (24, 0.7), (192, 0.2), (256, 0.9)$, $kw = 200, 800, 1600$, and $m = 1, 4, 6$. Seven common planes $\tau = -1, -0.5, 0, 0.5, 1, 1.5, 2$ are sampled. The electric longitudinal fraction is $\int_{\text{ROI}} |E_z|^2 / \int_{\text{ROI}} (S_0 + |E_z|^2)$. Maximum local fractions use declared bright-pixel masks $S_0 > 10^{-9} S_{0,\text{max}}$ and $10^{-6} S_{0,\text{max}}$. Full-plane retained transverse power, Stokes imbalance, and flux are reported separately from ROI quantities. The retained transverse power agrees with the incident scalar normalization to approximately 10^{-9} in the checked cases. Apparent remote FFTLog wrap tails are not interpreted as physical grazing-angle content.

At $R_0 = 256$, $\eta = 0.9$, $kw = 800$, the largest sampled relative ROI axial-flux correction over the three orders is approximately 1.91×10^{-4} . The sampled maxima of the longitudinal electric fraction across the three physical cases rise from 3.37×10^{-5} – 3.70×10^{-4} at $kw = 800$ to 5.40×10^{-4} – 5.77×10^{-3} at $kw = 200$. Small values in these cases do not establish a universal bound near dark pixels, high orders, or tighter focusing.

Independent Cartesian spectral integration enforces $\mathbf{k} \cdot \tilde{\mathbf{E}} = 0$ and $Z_0 \tilde{\mathbf{H}} = \mathbf{k} \times \tilde{\mathbf{E}}/k$ directly. Angular resolutions 64,128,256 agree with the radial recurrence formulas at matched quadrature points to relative E, H error below 1.4×10^{-12} and relative E_z error below 6.3×10^{-12} . Comparison to stored interpolated radial fields instead gives errors up to 1.48×10^{-7} , which include interpolation and differing spectral increments. Operator verification and grid verification are therefore reported separately.

12 Input mismatch and two-dimensional controls

The input-tolerance study uses $R_0 = 24, 96$, $\eta = 0.35, 0.7$, $m = 1, 2, 4$ on $[-8, 8]$. Signed power imbalance weights the two channels by $\sqrt{1 + \delta}$ and $\sqrt{1 - \delta}$, with $\delta = \pm 0.005, \pm 0.01, \pm 0.02, \pm 0.05$. Differential radial scales use $f(s/(1 \pm \delta/2))$ with $\delta = \pm 0.01, \pm 0.02$. Differential ring radii use $R_{\pm} = R_0(1 \pm \delta/2)$ with $\delta = \pm 0.005, \pm 0.01$; w and α remain fixed. Each distorted radial channel is normalized back to its specified input power, so power imbalance is not silently added to a geometric mismatch.

Differential quadratic phase is imposed as $\exp[\pm i\beta(s/R_0)^2/2]$ with $\beta = \pm 0.1, \pm 0.3$ rad. This is a spatial phase error at the input, distinct from a uniform post-propagation channel phase. Common scale changes 0.99, 1.01, a common quadratic phase, and source apertures at 1.5, 2, 3 times r_0 are controls. The finite aperture uses a raised-cosine transition of width $0.5w$; transmitted input power is recalculated and retained. All events use the nominal proxy aperture. This protocol does not claim to track a deformed actual-zero contour for a nonaxisymmetric beam.

The 324 beam cases yield 315 defined selected peaks and nine cases without a peak in the search interval, all involving the smallest source aperture. Six further selected peaks lack a preceding half-height crossing. In total, 309 individual half-heights and 304 paired advances are defined; five additional differences lack the reference crossing. Missing quantities retain separate peak, crossing, and paired-event statuses. The

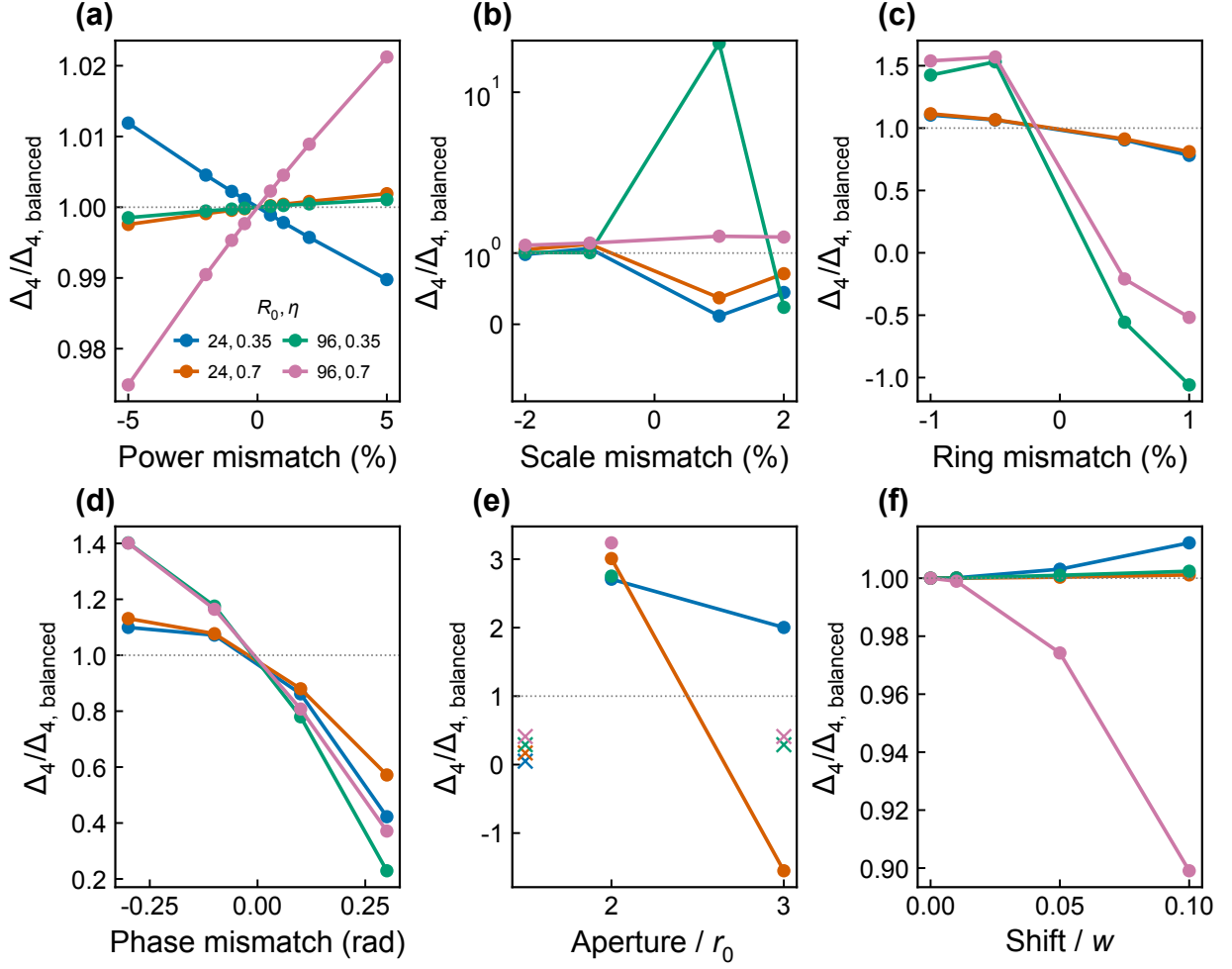


Figure TA3: Input sensitivity of the nominal-proxy $m = 4$ advance, divided by the corresponding balanced-input value. The legend gives R_0, η . (a) Signed power imbalance. (b) Differential radial scale; a symmetric logarithmic vertical axis retains large branch-dependent changes. (c) Differential ring radius. (d) Differential quadratic phase at r_0 . (e) Finite source aperture; crosses denote undefined paired advances and their vertical positions carry no numerical meaning. (f) Displacement of the plus channel in a two-dimensional calculation. These perturbation levels are tests, not accepted hardware tolerances.

large change at $R_0 = 96, \eta = 0.35$, positive radial scale mismatch is retained together with its complete extrema list; it cannot be interpreted as a small linear tolerance coefficient.

Finite source apertures can produce very weak early lobes, which remain eligible under the declared first-peak rule. For $R_0 = 24, \eta = 0.7$ and a source aperture $3r_0$, the selected $m = 1, 4$ peaks occur near $\tau = -6.11457, -5.68705$, well before the largest caustic peak. A three-level source, radial, and axial refinement changes their last half-heights by less than 1.3×10^{-8} and their difference by less than 2.3×10^{-8} . The $m = 2$ preceding half-height remains missing in that search window. Thus the large aperture-induced changes are not interpreted as small shifts of a uniquely tracked caustic peak.

Translations are implemented by the translation invariance of the exact scalar propagator applied to cached complex channels. The offsets are $d/w = 0, 0.01, 0.05, 0.1$, for a shifted plus channel, opposite shifts $\pm d/2$, or a common shift $d/2$. Nominal circular-aperture integration uses polar quadrature with a 64,128,256-level check. Opposite half shifts and a common half shift have equal circular-aperture channel integrals by rotational symmetry, although their two-dimensional polarization textures differ. This distinction prevents identical integrated values from being mistaken for identical fields.

Elliptical inputs at $R_0 = 24, \eta = 0.7, m = 1, 4$ use the area-preserving map $(x, y) \mapsto (x/a, ay)$, $a = \exp(e/2)$, with $e = 0.005, 0.01$. Both the envelope and vortex azimuth are transformed. A signed

angular Fourier expansion is propagated with the Fresnel kernel; mode offsets $\pm 16, \pm 24, \pm 32$ are checked. At $e = 0.01$, the omitted incident modal power outside the widest range is below 8.3×10^{-14} , and the event values agree across the last two mode ranges at saved precision. Direct two-dimensional Fresnel quadrature at three field points, independently refining source step and angular samples from 256 to 512, agrees with the stored modal field to relative channel error below 9.2×10^{-10} , including radial interpolation.

For common ellipticity $e = 0.005$, the $m = 4$ advance is 0.3120567; at $e = 0.01$ it is 0.5545883, versus approximately 0.291969 for the circular input. Deforming only the plus channel at $e = 0.01$ gives 0.4195528. Rotating the second ellipse by 90° leaves its circular-aperture intensity integral unchanged but changes its field map. These results illustrate why a calibrated input shape is needed even when total channel powers are balanced.

13 Synthetic measurement from circular-channel counts

The original detector study uses the computed complex fields at $R_0 = 24, 96$, $\eta = 0.35, 0.7$, and $m = 1, 2, 4$; the focused diagnosis below adds $m = 3$ without replacing the original records. The true circular-channel intensities, $I_\pm = |U_{m\pm 1}|^2/2$, are normalized by the incident transverse power. These are synthetic conditions; no camera, translation stage, or experimental performance is inferred from them. Unit detection efficiency is assumed in the reference count N per axial plane. A different known efficiency can be absorbed into N .

Before counting, each intensity is convolved with a circular Gaussian point-spread function. For a radial intensity the convolution is evaluated as

$$I_\sigma(r) = \int_0^\infty I(s) \frac{s}{\sigma^2} e^{-(r-s)^2/(2\sigma^2)} I_{0e}(rs/\sigma^2) ds, \quad (\text{TA49})$$

where $I_{0e}(t) = e^{-t} I_0(t)$ is the scaled modified Bessel function. Each square detector pixel is then integrated using 3×3 Gauss–Legendre quadrature. The circular aperture contains pixels whose centres lie within the nominal proxy radius. It therefore includes finite pixel sampling of the aperture as well as averaging within each pixel. Tests with 5×5 and 7×7 quadrature changed the pilot half-height positions by less than 2×10^{-8} in τ . The radial convolution agreed with an independently known Gaussian-convolution solution to 2.93×10^{-9} in relative L^2 error.

Let $p_{\pm j}$ be the resulting fraction of incident power in pixel j . The count means, before background subtraction, are

$$\lambda_{+j} = N(1+g)[(1-a)p_{+j} + bp_{-j}] + B, \quad (\text{TA50})$$

$$\lambda_{-j} = N(1-g)[ap_{+j} + (1-b)p_{-j}] + B, \quad a = c + \delta/2, \quad b = c - \delta/2. \quad (\text{TA51})$$

Independent Poisson counts and additive Gaussian read noise are generated for the two channels. The baseline has $N = 10^6$ reference counts per plane, axial step 0.1 mm, pixel pitch $1 \mu\text{m}$, blur standard deviation $1 \mu\text{m}$, read-noise standard deviation one electron per pixel, and background $B = 0.1$ electron per pixel. The gain standard deviation is 0.1% with cross-order correlation 0.8. The two background calibration offsets have standard deviation 0.001 electron per pixel and correlation 0.5, and are shared across scans. The mean polarimetric mixing is $c = 0.002$, with shared asymmetry standard deviation 0.0002. The monitor has mean $100N$ and independent Poisson noise at each plane, shared across the compared orders.

The common axial offset has standard deviation 0.02 mm. Independent order-specific offsets have standard deviation 0.01 mm, and independent linear scan drifts have standard deviation 0.01 mm at the scan endpoints. For every realization, the intensity is evaluated at the displaced physical plane while the aperture remains at its nominal position. Shifting only the already integrated moving-aperture curve would describe a different measurement and is not used.

Since the aperture weights are binary, the sum of independent Poisson pixel counts is exactly Poisson with mean $\sum_j \lambda_{\pm j}$. We use this identity to simulate the aggregate channel counts, with read variance equal to the number of retained pixels times the per-pixel variance. This retains the count statistics of the full image

without drawing every pixel in every realization. Representative full raw channel images and aggregate count traces are retained in the data archive. Noise is never added as a prescribed percentage of the final Q .

The scan covers $-1.5 \leq \tau \leq 2.5$. A cubic Savitzky–Golay filter uses a nominal width 0.25 in τ , rounded to an odd number of samples with at least five points. The estimator selects the largest interior smoothed maximum within $0.2 \leq \tau \leq 1.8$, records all additional maxima, and uses a three-point quadratic peak value. The half-height event is the nearest preceding rising linear crossing. Negative counts are retained. A missing maximum or crossing is an extraction failure; returning a number does not imply that the correct physical peak was selected.

The linearized confidence interval propagates the full covariance through the smoothing matrix and the peak/crossing functional. It retains peak–crossing covariance, shared monitor and calibration errors, and cross-order axial errors. For this simulation benchmark the covariance is evaluated at the known expected detector signal. It is therefore an *expected-signal linearization*, not a claim that these parameters can be estimated without calibration in an experiment. Bias and coverage are reported separately against the ideal continuous-aperture event and the noise-free pixel-and-blur event. Unknown systematic gain bias is not absorbed into a random-error interval.

The design contains 96 scenarios with 1000 realizations each: marginal sweeps of $N = 10^4$ – 10^7 , axial steps 0.05–1 mm, gain standard deviation 0–1%, blur 0–5 μm , pixel pitch 0.5–4 μm , read noise 0–3 electrons, and independent axial offset 0–0.1 mm, together with two predefined interacting/systematic stress scenarios. These ranges define marginal sweeps rather than a full Cartesian product. The scenario table reports event statuses as source data. Four representative conditions were repeated independently with 4000 realizations. For $R_0 = 96$, $\eta = 0.7$, $m = 4$, the correct-sign fractions were 0.251 and 0.255 at $N = 10^6$, and 0.979 and 0.98075 at $N = 10^7$. These repetitions support the distinction between those two count regimes; they do not establish a universal detection threshold.

Table TA5: Baseline synthetic measurement for $m = 4$ relative to $m = 1$. Bias and interval coverage refer to the ideal continuous-aperture advance. Widths are the full linearized 95% interval; the correct-sign fraction is empirical.

R_0	η	Δz_4 (mm)	Bias (mm)	Width (mm)	Coverage	Correct sign
24	0.35	3.69334	−0.00314	0.2363	0.957	1.000
24	0.7	3.78726	−0.02302	0.5604	0.950	1.000
96	0.35	0.45746	−0.01091	0.6265	0.955	0.994
96	0.7	0.48972	−1.71029	8.3827	0.915	0.251

For the baseline at $R_0 = 24$, both tested non-reference orders have the correct advance sign in all 1000 realizations. At $R_0 = 96$, $\eta = 0.7$, however, the ideal $m = 4$ advance is 0.48972 mm and the estimated mean is biased by −1.71029 mm. Only 25.1% of realizations have the correct sign. The failure is associated with low local counts and peak selection under noise, rather than the absence of a mathematical crossing in the ideal signal. In the combined low-count/coarse-image stress case, 2.5% of paired estimates cannot be extracted, while the sign fraction among extracted estimates is approximately 0.502. Very wide linearized intervals in this regime are not evidence of a useful measurement.

13.1 Four-stage failure diagnosis and frozen estimator comparison

Before drawing the focused random samples, we fixed three extraction rules: (i) the retained width-0.25 τ blind largest-peak estimator; (ii) a known-peak-location-assisted diagnostic using the same smoothing but selecting the local maximum nearest the known noiseless detector peak; and (iii) one implementable blind alternative using a pre-fixed width 0.75 τ and the largest peak. The third rule uses neither the true displacement nor D_m . One thousand calibration replicates determine the covariance, and 3000 independent validation replicates determine bias, extraction, interval coverage, and sign recovery. The protocol and seeds are stored before the validation draws.

Four stages isolate the failure. A is the continuous-field event in the specified ROI. B applies pixel integration and blur without random noise. C applies the fixed smoothing and peak/crossing routine to the

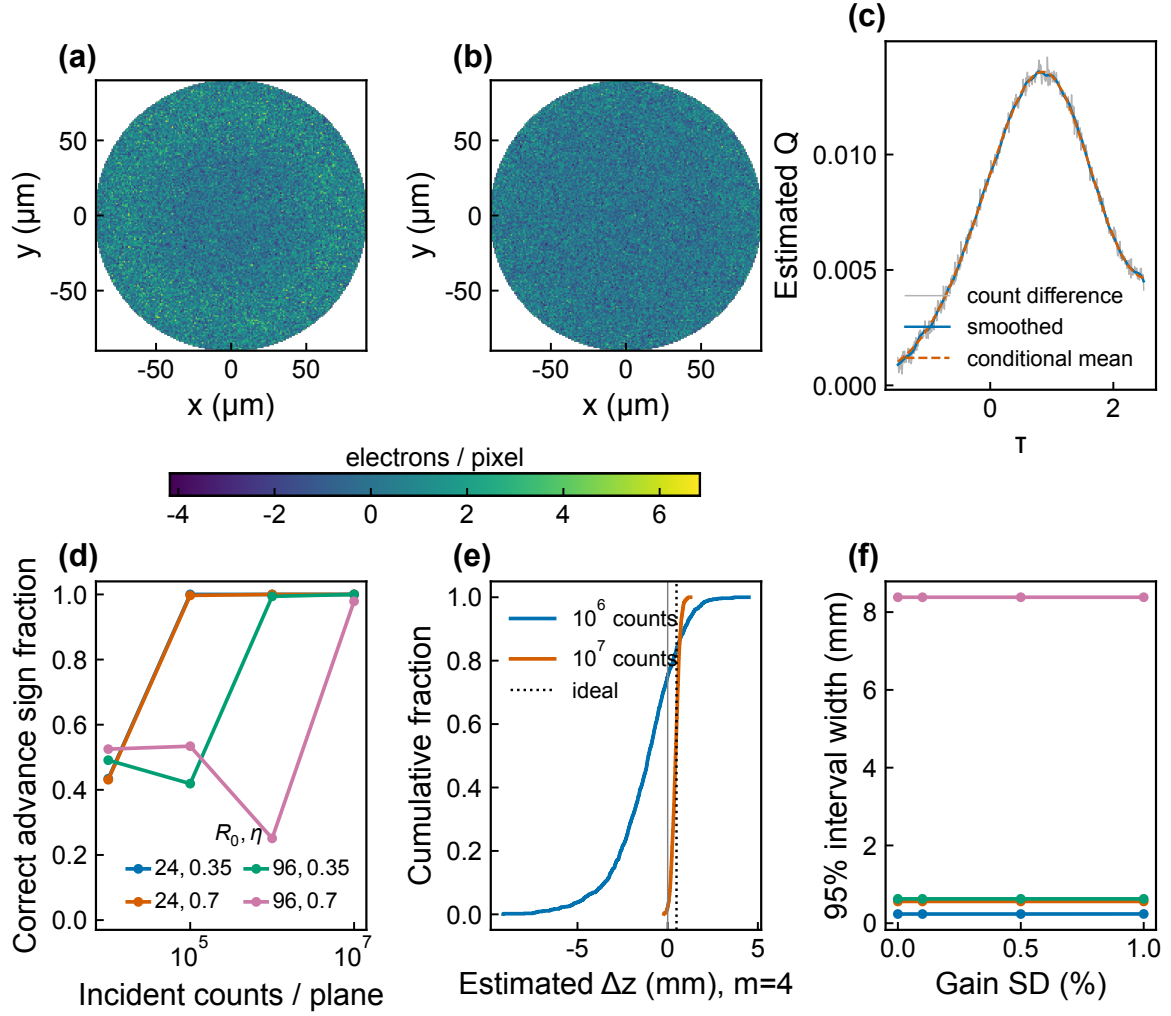


Figure TA4: Synthetic count-level measurement. (a,b) Independently generated raw circular-channel images at $\tau = 0$, $R_0 = 24$, $\eta = 0.7$, $m = 4$, with the same electron-count scale. (c) One aggregate count trace, its smoothed estimate, and the conditional mean. (d) Correct-sign fraction for Δz_4 in the count sweep with other conditions fixed. (e) Full empirical cumulative distributions at $R_0 = 96$, $\eta = 0.7$; the dotted line is the ideal continuous-aperture advance. (f) Linearized interval widths with the stated gain covariance. All quantities are simulated, and each sweep point contains 1000 realizations.

noiseless detector curve. D applies the same routine to the raw random channel counts. For the difficult $R_0 = 96$, $\eta = 0.7$, $m = 4$ case, A–D give 0.489722, 0.484247, 0.484237, and -1.242756 mm for the retained blind method at $N = 10^6$. The imaging bias is -0.005475 mm and the deterministic estimator bias is -0.000010 mm, whereas the random bias from C to D is -1.726993 mm. The original 1000- and 4000-realization results are therefore corroborated without being overwritten.

The estimator returns a number in every focused validation scan, so all-run and extracted-only fractions coincide here. At $N = 10^6$, the retained blind and location-assisted rules select the same discrete local maximum in only 14.9%, 13.6%, and 13.0% of the $m = 2, 3, 4$ scans. Equality of these sample indices does not establish that either selected the same physical peak branch. The retained rule gives correct-sign fractions of 42.0%, 32.0%, and 25.7%; the location-assisted diagnostic gives only 47.4%, 42.7%, and 38.4%. This diagnostic uses the noiseless peak location but leaves the noisy peak value, physical branch, and rising crossing free, so its residual error cannot be attributed uniquely to the crossing. The wider blind rule improves the fractions to 50.9%, 52.5%, and 54.9%, still insufficient for a feasibility claim.

At the detector half-height and $N = 10^6$, the expected (I_+ , I_-) ROI counts are (85.81, 6.83) for $m = 1$ and (163.44, 29.75) for $m = 4$. The aperture contains hundreds to thousands of pixels depending on order, so one electron of independent read noise per pixel accumulates in the aggregate. A Poisson-only ablation gives retained-estimator sign fractions 68.9%, 85.1%, and 94.0% for $m = 2, 3, 4$. Counting plus read noise reduces them to 43.5%, 37.1%, and 26.9%, reproducing most of the full failure; counting plus background alone gives 57.5%, 65.2%, and 67.3%. Hence the dominant modeled cause is accumulated read noise, with peak/crossing selection converting it into order-dependent bias; background adds degradation. It is not accurately described as incident photon shortage alone.

At $N = 10^7$, the retained estimator gives correct signs in 74.6%, 91.9%, and 97.6% of the $m = 2, 3, 4$ scans, and the pre-fixed wider rule gives 86.5%, 99.1%, and 99.9%. The smaller $m = 2$ shift remains more difficult. At $R_0 = 24$ and $N = 10^6$, all $m = 2, 3, 4$ signs are recovered for both tested apodizations under the retained estimator. These outcomes are conditional on the stated synthetic detector and do not set a general photon or hardware threshold.

Table TA6: Four-stage diagnosis at $R_0 = 96$, $\eta = 0.7$, and $N = 10^6$ for the retained blind estimator. A is the continuous-field event, B includes noiseless pixel and blur response, C applies the fixed smoothing and peak rule to B, and D is the mean over 3000 independent count-level validation scans. Values are millimeters except the final percentage.

m	A	B	C	D	B–A	C–B	D–C	sign (%)
2	0.149997	0.137186	0.137255	−0.304859	−0.012811	+0.000069	−0.442114	42.0
3	0.314388	0.311861	0.311892	−0.715232	−0.002527	+0.000030	−1.027124	32.0
4	0.489722	0.484247	0.484237	−1.242756	−0.005475	−0.000010	−1.726993	25.7

Table TA7: Estimator comparison for the difficult $R_0 = 96$, $\eta = 0.7$ condition. Sign percentages and random-mean advances use 3000 independent validation scans. The location-assisted diagnostic selects the noisy local maximum nearest the known noiseless peak location; it does not fix the noisy peak value, physical branch, or half-height crossing. The wide estimator is a pre-frozen blind rule.

N	estimator	correct sign (%)			mean advance (mm)		
		$m = 2$	$m = 3$	$m = 4$	$m = 2$	$m = 3$	$m = 4$
10^6	current	42.0	32.0	25.7	−0.305	−0.715	−1.243
10^6	location assisted	47.4	42.7	38.4	−0.119	−0.315	−0.590
10^6	wide	50.9	52.5	54.9	0.048	0.071	0.135
10^7	current	74.6	91.9	97.6	0.126	0.289	0.449
10^7	location assisted	73.3	91.4	97.7	0.130	0.302	0.473
10^7	wide	86.5	99.1	99.9	0.132	0.303	0.473

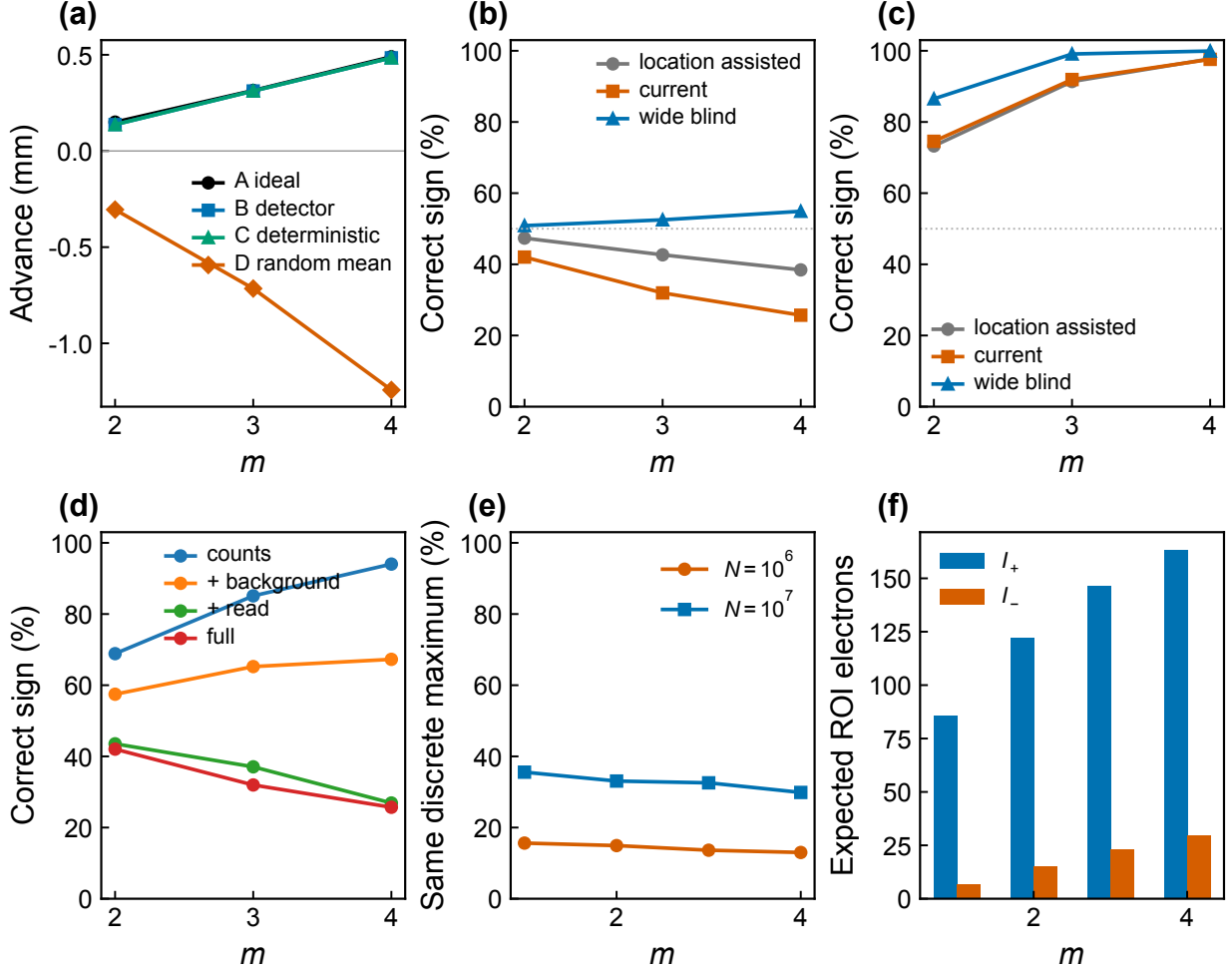


Figure TA5: Focused diagnosis of the difficult $R_0 = 96$, $\eta = 0.7$ measurement. (a) Continuous-field (A), noiseless detector (B), deterministic estimator (C), and random validation mean (D) for the retained blind rule. (b,c) Correct-sign fractions for the location-assisted diagnostic, retained blind, and pre-fixed wider blind estimators at 10^6 and 10^7 reference counts per plane. (d) Noise-source ablations at 10^6 counts for the retained estimator. (e) Fraction for which the retained blind and location-assisted rules select the same discrete local maximum; this is not a physical peak-identity label. (f) Expected circular-channel ROI counts at the noiseless detector half-height for 10^6 incident reference counts. Each focused fraction uses 3000 independent validation scans; the assisted rule is diagnostic and the wider blind rule was fixed before those scans.

13.2 Proposed channel-resolved optical implementation

Figure TA6 maps the modeled observable to a laboratory measurement. A linearly polarized continuous-wave source at 632.8 nm is spatially filtered and expanded before a polarization Mach–Zehnder interferometer. The two arms use phase-only spatial-light-modulator holograms with Fourier filtering to encode the same finite-energy circular Airy envelope $f_A(r)$ and the charges $q_+ = m - 1$ and $q_- = m + 1$. A half-wave plate and the hologram efficiencies set the incident channel powers. The arms are recombined in orthogonal linear polarizations; a quarter-wave plate at 45° converts them to the circular basis used in the main text. A uniform inter-arm phase changes the polarization orientation but not $I_\pm$ or S_3 . Transverse registration, equal radial scale, and power balance must nevertheless be calibrated because they change the measured channel intensities.

The acquisition head is translated through the autofocusing region, or an equivalent translated relay images each plane. A quarter-wave plate followed by a polarizing beam splitter sends the two circular components to synchronized detectors. A monitor photodiode records the incident normalization. Full frames are retained, so the scaled proxy boundary is applied digitally with physical radius

$$\rho_s(z) = \frac{2kw^3x_s}{z}, \quad x_s = j'_{m,1}, \quad (\text{TA52})$$

rather than with a moving mechanical iris. After dark, flat-field, gain, polarimetric-leakage, and image-registration corrections, let $\tilde{C}_{\pm j}(z)$ denote the corrected count in pixel j . The directly measured traces are

$$\hat{Q}(z) = \frac{1}{\hat{P}_{\text{in}}(z)} \sum_{j \in \text{ROI}(z)} [\tilde{C}_{+j}(z) - \tilde{C}_{-j}(z)], \quad (\text{TA53})$$

$$\hat{T}(z) = \frac{1}{\hat{P}_{\text{in}}(z)} \sum_{j \in \text{ROI}(z)} [\tilde{C}_{+j}(z) + \tilde{C}_{-j}(z)], \quad \hat{\bar{P}}_3(z) = \frac{\hat{Q}(z)}{\hat{T}(z)}. \quad (\text{TA54})$$

The same fixed peak and rising-crossing rules used in the count simulation then give $z_{50}(m)$ and $\Delta z_m = z_{50}(1) - z_{50}(m)$. For the alternative actual-zero analysis, the first positive-to-negative zero is extracted from the calibrated, azimuthally averaged S_3 map at each plane and its branch status is retained. The axial scan increment is only the sampling interval; repeat scans and the calibrated covariance are required to determine event uncertainty.

Input certification is kept separate from this primary acquisition. A quarter-wave plate and rotating linear polarizer provide full-Stokes images, whereas interference of each circularly selected component with a matched spherical reference checks q_+ and q_- . A linear-analyzer petal pattern is not used to infer m , because the fixed $P = 1$ family has the same twofold modulation for every m . These modules follow established SLM preparation, Stokes-polarimetry, and circular-channel interferometry procedures [1–3]. Figure TA6 is an implementable proposal; the present work has not built this apparatus, and the feasibility numbers above remain predictions of the stated detector model.

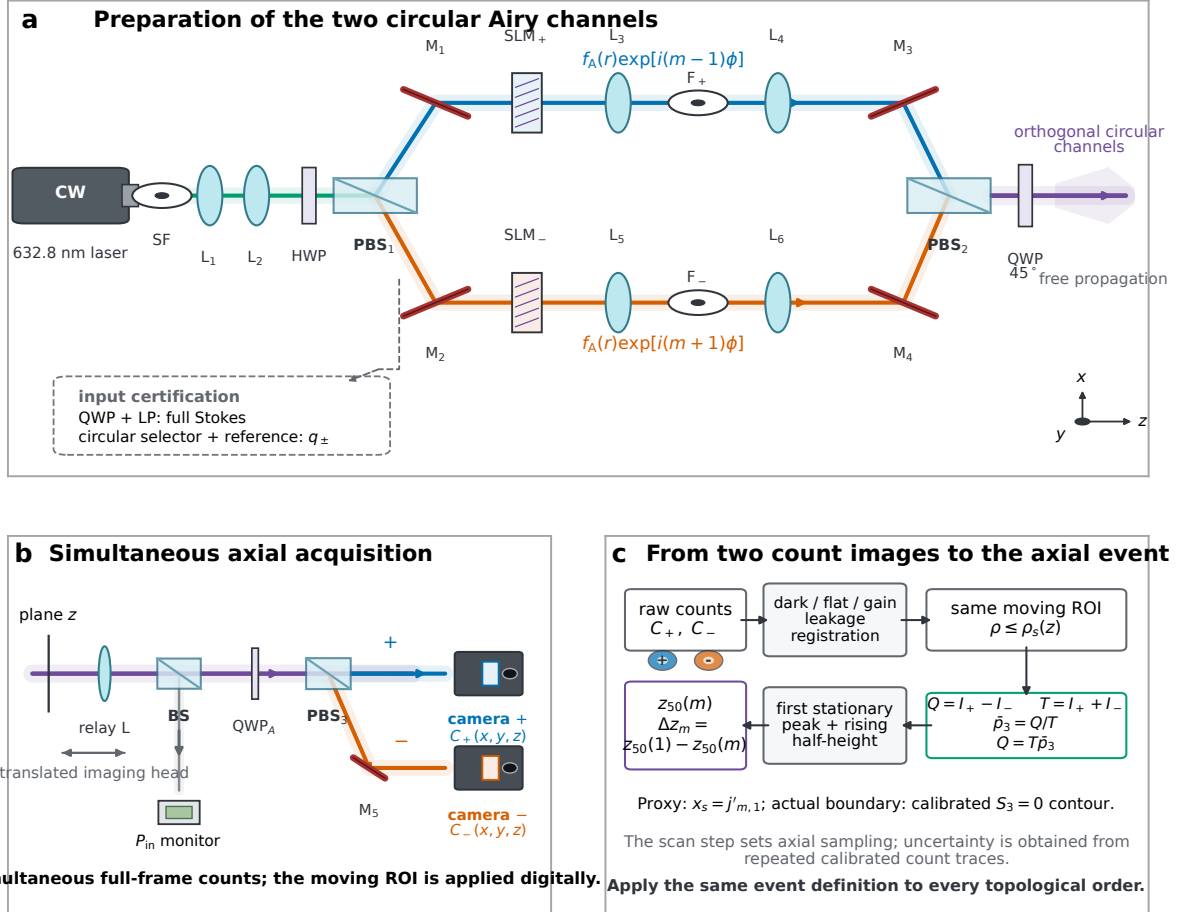


Figure TA6: Proposed measurement of the topological-order-dependent axial event. (a) A polarization Mach–Zehnder interferometer prepares the same circular Airy envelope in the $q_+ = m - 1$ and $q_- = m + 1$ arms and converts the recombined field to orthogonal circular polarizations. The dashed branch contains independent input diagnostics. SF, spatial filter; L1–L2, beam expander; HWP, half-wave plate; PBS, polarizing beam splitter; SLM, spatial light modulator; QWP, quarter-wave plate. (b) A translated imaging head separates and records the two circular-channel count images simultaneously while a photodiode monitors incident power. (c) Calibrated full-frame counts are integrated in the digital moving ROI to obtain Q , T , and $\bar{\rho}_3$ before the fixed peak and rising-half-height estimator is applied. This is a proposed optical implementation, not an experiment performed in the present work.

14 Finite radius residual mechanism

The Airy convolution

For $\alpha > 0$ define the one-dimensional Fresnel propagation of the apodized Airy function:

$$F(y, \zeta) = \frac{1}{\sqrt{2\pi i \zeta}} \int_{-\infty}^{\infty} \text{Ai}(t) e^{\alpha t} e^{i(y-t)^2/(2\zeta)} dt. \quad (\text{TA55})$$

The initial profile has Fourier transform $\exp[(\alpha - ik)^3/3]$. Multiplication by $\exp(-i\zeta k^2/2)$ and completion of the cubic phase give

$$F(y, \zeta) = \text{Ai}(y - \zeta^2/4 + i\alpha\zeta) e^{\Phi(y, \zeta)}, \quad (\text{TA56})$$

$$\Phi(y, \zeta) = \alpha(y - \zeta^2/2) + i[\zeta(y + \alpha^2)/2 - \zeta^3/12]. \quad (\text{TA57})$$

This is the finite-energy Airy solution of Siviloglou and Christodoulides, Eq. (5) [4]. Setting $t = R_0 - s$, $a = R \cos \theta$ and $y = R_0 - a$ converts the full-line source integral without its factor s to $\sqrt{2\pi i \zeta} e^{-ia^2/(2\zeta)} F(R_0 - a, \zeta)$. Applying $i\zeta \partial_a$ restores the factor s . The sign follows directly by differentiating $\exp(-ias/\zeta)$.

For integer q , the Bessel angular identity [5] is

$$J_q(v) = \frac{i^q}{2\pi} \int_0^{2\pi} e^{-iv \cos \theta} e^{iq\theta} d\theta. \quad (\text{TA58})$$

Let $X = R_0 - \zeta^2/4 + i\alpha\zeta$. After removing a nonzero common factor and normalizing by $\zeta^2/2$, the angular field is

$$\mathcal{H}_q = \frac{1}{2\pi} \int_0^{2\pi} e^{iq\theta} G(a, \zeta) \left[B(a, \zeta) \text{Ai}(X - a) - \frac{2i}{\zeta} \text{Ai}'(X - a) \right] d\theta, \quad (\text{TA59})$$

$$G(a, \zeta) = e^{-i\zeta a/2 - \alpha a - ia^2/(2\zeta)}, \quad B(a, \zeta) = 1 - \frac{2i\alpha}{\zeta} + \frac{2a}{\zeta^2}. \quad (\text{TA60})$$

The normalized proxy signal is therefore

$$\tilde{Q}_m = \frac{V e^{-2\eta\tau\epsilon - \eta\tau^2\epsilon^3/2}}{I_{0,m}} \int_0^{b_m} x \left(|\mathcal{H}_{m-1}|^2 - |\mathcal{H}_{m+1}|^2 \right) dx, \quad (\text{TA61})$$

where $\epsilon = R_0^{-1/2}$, $V = 1 + \tau\epsilon^2/2$ and $I_{0,m} = 2mJ_m^2(b_m)$. This normalization differs from Q_m by a positive factor independent of τ , so it leaves peaks and half-height events unchanged. The V and exponential factors must remain during event extraction.

For reduced R-S propagation, use d instead of ζ inside $\mathcal{H}_q$. Since d depends on x , its signal prefactor must be kept inside the x integral:

$$W_{\text{RS}}(x) = \frac{d\epsilon}{2} e^{-2\eta\tau\epsilon - \eta\tau^2\epsilon^3/2 - \alpha(R/kw)^2}, \quad (\text{TA62})$$

$$\tilde{Q}_m^{\text{RS}} = \frac{1}{I_{0,m}} \int_0^{b_m} x W_{\text{RS}}(x) \left(|\mathcal{H}_{m-1}(d)|^2 - |\mathcal{H}_{m+1}(d)|^2 \right) dx. \quad (\text{TA63})$$

This expression retains the exact geometry of the manuscript's reduced kernel. It is not a full Maxwell or unexpanded Rayleigh-Sommerfeld calculation. The angular identity sums the complete field coherently and introduces no arbitrary spectral partition.

Controlled negative-source extension

Replacing the physical interval $s \geq 0$ by the full real line adds an integral over $s = -v < 0$. Since $|J_q(t)| \leq 1$ for real t and integer q , the positive-axis Airy bound [6] and convexity of $(R_0 + v)^{3/2}$ give, for $\alpha < \sqrt{R_0}$,

$$|\delta I_q| \leq \frac{e^{\alpha R_0 - 2R_0^{3/2}/3}}{2\sqrt{\pi}R_0^{1/4}(\sqrt{R_0} - \alpha)^2}. \quad (\text{TA64})$$

Indeed, $\text{Ai}(u) \leq e^{-2u^{3/2}/3}/(2\sqrt{\pi}u^{1/4})$ for $u > 0$, and $2(R_0 + v)^{3/2}/3 \geq 2R_0^{3/2}/3 + \sqrt{R_0}v$. The remaining integral is $\int_0^\infty v e^{-(\sqrt{R_0} - \alpha)v} dv$. The resulting dimensionless source-integral bounds are 1.1541×10^{-36} at $R_0 = 24$, 4.8580×10^{-76} at 40 and 2.433×10^{-275} at 96. These are bounds before the known propagation prefactor, not estimates obtained from cancellation. This artificial extension cannot account for the observed 10^{-2} changes in $R_0\Delta$.

14.1 Coefficients and event recursion

The angular formula admits direct expansion around $\epsilon = 0$. In the Fresnel case,

$$a = \frac{\epsilon x \cos \theta}{V}, \quad X = -\tau + i\eta + \epsilon^2(-\tau^2/4 + i\eta\tau/2), \quad \zeta a/2 = x \cos \theta. \quad (\text{TA65})$$

Expand V^{-1} geometrically, the two remaining exponentials by power-series multiplication, and $\text{Ai}(X - a)$ about $-\tau + i\eta$. Its derivatives satisfy $A^{(k)} = XA^{(k-2)} + (k-2)A^{(k-3)}$ for $k \geq 3$, with $A'' = XA$. The field coefficients $H_{q,n}$ then determine every intensity coefficient by the convolution $\sum_{j=0}^n H_{q,j}H_{q,n-j}^*$. The common signal prefactor is convolved at the same total order. There is no truncation by derivative number followed by uncontrolled squaring.

Write $Q_m(\tau, \epsilon) = \sum_n q_{m,n}(\tau)\epsilon^n$, $p_m = p_0 + \sum_{n \geq 1} p_{m,n}\epsilon^n$ and $h_m = h_0 + \sum_{n \geq 1} h_{m,n}\epsilon^n$. For a simple peak and rising crossing, coefficient extraction gives

$$p_{m,n} = -\frac{[\epsilon^n] \partial_\tau Q_m(p_0 + \sum_{j < n} p_{m,j}\epsilon^j, \epsilon)}{q_{m,0}''(p_0)}, \quad (\text{TA66})$$

$$h_{m,n} = \frac{\frac{1}{2}[\epsilon^n] Q_m(p_0 + \sum_{j < n} p_{m,j}\epsilon^j, \epsilon) - [\epsilon^n] Q_m(h_0 + \sum_{j < n} h_{m,j}\epsilon^j, \epsilon)}{q_{m,0}'(h_0)}. \quad (\text{TA67})$$

The unknown $p_{m,n}$ does not enter the peak height at this order because $q_{m,0}'(p_0) = 0$. This proves the recursion under its nondegeneracy assumptions. The coefficients $c_n = h_{1,n} - h_{3,n}$ are fixed by the model and event definition. They do not depend on a chosen set of propagation radii. The normalized full-line signal is analytic locally in ϵ where $V \neq 0$; the implicit function theorem supplies local analytic events while their denominators remain nonzero. The source extension contributes a beyond-all-orders term to the physical half-line integral. These local facts alone do not certify a global convergence radius for the event series.

Table TA8: Analytic event-difference coefficients for $m = 3$ relative to $m = 1$ at $\eta = 0.35$, with $\Delta = \sum_n c_n \epsilon^n$ and $\epsilon = R_0^{-1/2}$. The first-order coefficient vanishes analytically.

n	c_n	n	c_n
2	4.41467828893	3	-1.44016807989
4	2.58765834920	5	3.36680299947
6	13.3529860411	7	21.1128977205
8	-49.8888463954	9	91.5175673968
10	237.809936151	11	-385.651361911
12	738.534340134	13	-809.692872335
14	3130.68031359	1	0 analytically

The computed c_1 is 1.2×10^{-15} . The first three nonzero coefficients reproduce the independent earlier fold derivation. Changing the Cauchy derivative-contour radius from 0.7 to 0.9 changes no coefficient through order 14 by more than 8.3×10^{-10} . Separate order-10 checks refine the radial and angular quadratures from 32/64 to 48/96 nodes. Cauchy extraction evaluates derivatives of an analytic formula; it does not fit propagated event values.

14.2 Why the minimum was missed

The scaled event series is $R_0\Delta = \sum_{n \geq 2} c_n R_0^{1-n/2}$. At fourth order it contains only $A_3 + B_3/\sqrt{R_0} + C_3/R_0$. Its stationary point is $(2C_3/|B_3|)^2 = 12.91361$, outside the observed intermediate-radius minimum. Adding the positive fifth- and sixth-order terms shifts the minimum toward the full calculation.

Table TA9: Minimum of the scaled advance $R_0\Delta$ for successive event-series truncations and the full propagation models at $m = 3$, $\eta = 0.35$. Truncation order denotes the power of ϵ .

Event truncation	Predicted minimum R_0	Minimum $R_0\Delta$
4	12.91361	4.21429595
5	24.97050	4.25708537
6	34.41707	4.27232531
8	35.78792	4.27406412
10	36.41148	4.27452784
12	36.37256	4.27450296
14	36.37128	4.27450221
Full Fresnel	36.37127	4.27450221
Reduced R-S	36.37130	4.27450539

These are order controls without coefficient fitting. The high orders do not merely improve a regression statistic: they predict the location and depth of the minimum and the sign of its curvature. The decomposition below additionally identifies which operations generate the missing terms.

For each order n , split the new signal coefficient into (i) $H_0 H_n^* + H_n H_0^*$, (ii) lower-field products and the common prefactor, and (iii) inherited nonlinear terms from the two implicit event equations. Applying the linear crossing functional $[q_n(p_0)/2 - q_n(h_0)]/q_0'(h_0)$ to (i) and (ii), and assigning the remainder of h_n to (iii), gives the following exact algebraic grouping in this declared expansion convention.

Table TA10: Signed contributions to the fifth- and sixth-order event coefficients at $m = 3$, $\eta = 0.35$, using the field-product and implicit-event grouping defined in the text.

Order	New field term	Lower products and prefactor	Nonlinear event terms	Total c_n
5	1.04031657	23.80479484	-21.47830841	3.36680300
6	22.98782712	54.38661509	-64.02145617	13.35298604

The large signed cancellations rule out interpreting c_5 or c_6 as the power carried by a separate branch. They are coherent finite-radius field corrections processed through peak normalization and a moving half-height. As an event control, forcing the target's normalization peak to the reference peak changes $R_0\Delta$ to 4.52744, 4.42926 and 4.36945 at 24, 40 and 96; the dip over these three points disappears. This artificial control isolates the importance of the peak condition but is not proposed as a replacement observable. Removing c_5 or c_6 from the fixed analytic series likewise displaces the minimum; all values are retained in `coefficient_decomposition.json`.

14.3 Predictions and independent numerical evidence

All coefficients were generated solely from the analytic identity; the generators never load propagation data. Predictions at the new radii were evaluated with these fixed coefficients. New controls are $R_0 = 20, 28, 36, 44, 48, 64, 80$; the report also recomputes the original radii and extends to 256. The maximum fourteenth-order error in $R_0\Delta$ against the full Fresnel identity on the new controls is 2.84×10^{-7} , attained at 20. Over sampled $24 \leq R_0 \leq 96$ it is below 7.1×10^{-8} . The small reduced R-S/Fresnel difference is retained separately, rather than absorbed into a fitted high-order term.

Table TA11: Scaled reduced R-S advance, fourteenth-order Fresnel event-series error, and reduced R-S minus Fresnel difference at $m = 3$, $\eta = 0.35$. All advances are relative to $m = 1$.

R_0	Reduced R-S $R_0\Delta$	Fourteenth-order Fresnel error in $R_0\Delta$	R-S minus Fresnel
24	4.286137374	$+7.02 \times 10^{-8}$	3.225×10^{-6}
36	4.274510642	-2.94×10^{-9}	3.178×10^{-6}
40	4.274925216	-2.94×10^{-9}	3.172×10^{-6}
48	4.277668295	-1.75×10^{-9}	3.166×10^{-6}
64	4.285437979	-4.95×10^{-10}	3.161×10^{-6}
96	4.299868386	-5.99×10^{-11}	3.161×10^{-6}

The angular implementation uses Gauss-Legendre integration in x and periodic trapezoidal integration in angle. The independent positive-source implementation evaluates the original oscillatory radial integral by composite Gauss-Legendre quadrature, with exact reduced R-S geometry and without a spectral transform, phase expansion, radial interpolation or axial spline. Separate tests halve the source panels and extend the exponential cutoff. Refining panels from 0.5 to 0.25 and cutoff from 22 to 26 changes the scaled advance by at most 4.64×10^{-10} . The finest direct-source results differ from the compact reduced R-S evaluation by at most 1.41×10^{-12} in $R_0\Delta$. The arbitrary precision implementation uses a third angular evaluation: an Airy Taylor series in a and analytic angular moments expressed as Bessel derivatives. It uses `mpmath` at 35 and 50 decimal digits, 20/28 radial nodes and 36/48 angular-series terms. At the two event locations for each of $m = 1, 3$ and $R_0 = 24, 40, 96$, the maximum signal difference between these precision settings is 6.0×10^{-33} ; the double-precision angular implementation differs by at most 1.45×10^{-15} . These are measured numerical differences, not interval-certified bounds.

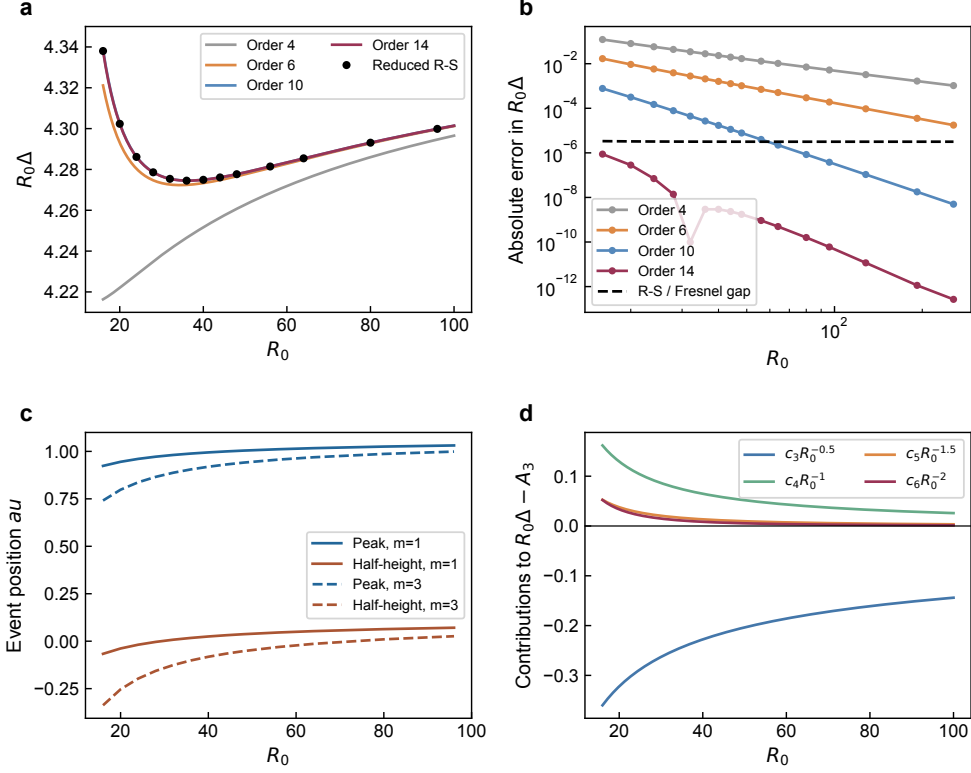


Figure TA7: Mechanism evidence. (a) Non-fitted event expansions and full reduced R-S values. (b) Truncation errors against the full Fresnel identity and the separately retained geometry correction. (c) First peak and rising half-height continuation. (d) Signed lower-order contributions to the scaled advance.

The first peaks and preceding rising crossings continue smoothly across the 16 tested radii from 16 to 256. The minimum crossing slope of the normalized signal is 0.1035; the minimum magnitude of the selected peak curvature is 0.2561. All detected later maxima are retained in `event_continuation.csv`. The selected first maxima remain separated from them. Thus the tested minimum is not associated with a detected event switch, peak degeneracy or a selected-zero jump.

14.4 Error accounting and interpretation

The signed residual admits the exact bookkeeping identity

$$\Delta_{RS} - \Delta_4 = (\Delta_{RS} - \Delta_F) + \sum_{n=5}^{14} c_n R_0^{-n/2} + (\Delta_F - \Delta_{14}). \quad (\text{TA68})$$

The first term is the model geometry correction, the second is the analytic high-order event contribution and the third is the retained series remainder. The source-extension bound and numerical evaluation error are reported separately. This identity closes the residual without allocating coherent powers to arbitrary spectral windows.

Table TA12: Source-extension, numerical, finite-series, and model-geometry contributions to the residual analysis for $m = 3$, $\eta = 0.35$. Numerical differences and the analytic bound are reported separately.

Contribution	Evidence and scale	Interpretation
Source extension	Analytic bound 1.16×10^{-36} on I_q at 24	Rigorous positive-source extension control
Angular and radial quadratures	48/96 to 72/128 nodes: scaled-advance change below 6.2×10^{-14}	Measured numerical convergence
Independent source quadrature	Source refinement change below 4.64×10^{-10} in scaled advance	Direct check of the original integral
Arbitrary precision signal evaluation	Maximum double/50-digit difference 1.45×10^{-15}	Independent Airy/Bessel implementation
Fourteenth-order remainder	Below 7.1×10^{-8} in sampled $R_0\Delta$, 24–96	Finite-order approximation, not quadrature noise
Reduced R-S versus Fresnel	$3.16\text{--}3.23 \times 10^{-6}$ in $R_0\Delta$, 24–96	Separate kernel geometry correction
Observed dip and recovery	24 to 40: -0.01121216 ; 40 to 96: $+0.02494317$	Resolved finite-radius behavior

For the reduced R-S model and fixed-proxy half-height event, the independently derived higher-order terms quantitatively account for the minimum. The peak and crossing branches remain continuous over the sampled radii. A decomposition into inward, outward, and endpoint powers remains representation dependent.

References

- [1] Mingze Liu, Pengcheng Huo, Wenqi Zhu, Cheng Zhang, Si Zhang, Maowen Song, Song Zhang, Qianwei Zhou, Lu Chen, Henri J. Lezec, Amit Agrawal, Yanqing Lu, and Ting Xu. Broadband generation of perfect Poincaré beams via dielectric spin-multiplexed metasurface. *Nature Communications*, 12: 2230, 2021. doi: 10.1038/s41467-021-22462-z.
- [2] Xin Yan, Xin Tao, Minghao Guo, Chunliang Zhou, Jingzhao Chen, Guanyu Shang, and Peng Li. Vector vortex beams: Theory, generation, and detection of Laguerre–Gaussian and Bessel–Gaussian types. *Photonics*, 12:1029, 2025. doi: 10.3390/photonics12101029.
- [3] Liu-Hao Zhu, Xiao-He Zhang, Xiang Zang, Guang-Hao Rui, Jun He, Chun-Xiang Xu, and Bing Gu. Diverse particle manipulation within different 2D planes via orientation-selective Airy vortex beams. *Journal of Applied Physics*, 137:223105, 2025. doi: 10.1063/5.0270605.
4. G. A. Siviloglou and D. N. Christodoulides, *Opt. Lett.* 32, 979–981 (2007), doi: 10.1364/OL.32.000979.
5. NIST DLMF, Sections 9.5 and 10.9.2, <https://dlmf.nist.gov/10.9.E2>.
6. NIST DLMF, Equation 9.7.15, <https://dlmf.nist.gov/9.7.E15>.